On the Predictive Skill of Artificial Intelligence-based Weather Models for Extreme Events using Uncertainty Quantification

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Abstract

Accurate prediction of extreme weather events remains a major challenge for artificial intelligence–based weather prediction systems. While deterministic models such as FuXi, GraphCast, and SFNO have achieved competitive forecast skill relative to numerical weather prediction, their ability to represent uncertainty and capture extremes is still limited. This study investigates how state-of-the-art deterministic artificial intelligence-based models respond to initial-condition perturbations and evaluates the resulting ensembles in forecasting extremes. Using four perturbation strategies (Gaussian, Perlin noise, Hemispheric Centered Bred Vectors, and Huge Ensembles), we generate 50 member ensembles for the August 2022 Pakistan floods and China heatwave, and complement these case studies with a global threshold-based evaluation. Ensemble skill is assessed against ERA5 and compared with IFS ENS and the AIFS ENS probabilistic model using deterministic and probabilistic metrics. Results show that simpler perturbations like Gaussian and Perlin noise produce similarly realistic ensemble spread and probabilistic skill as flow-based approaches like HCBV and HENS, narrowing but not closing the performance gap with numerical weather prediction ensembles, or native probabilistic models which retain the highest probabilistic skill across variables. Model choice is the dominant factor for ensemble performance, not perturbation method. Across variables, models capture temperature extremes more effectively than precipitation. These findings demonstrate that simple input perturbations can extend deterministic models toward probabilistic forecasting in hardware-constrained settings, supporting artificial intelligence-driven early warning systems.

1 Introduction

Artificial Intelligence (AI) is rapidly reshaping the field of weather and climate forecasting, including the assessment and management of natural hazard risks [Camps-Valls et al., 2025]. Traditionally, Numerical Weather Prediction (NWP) relies on solving complex systems of partial differential equations to simulate atmospheric processes. However, this approach demands substantial computational resources, which constrains both spatial and temporal resolution. Given the inherently chaotic nature of the atmosphere, it is also crucial to account for uncertainties in both the initial conditions and the model itself. Ensemble forecasting addresses this challenge by generating multiple simulations to represent a range of possible outcomes, forming the basis for probabilistic forecasts [Leutbecher and Palmer, 2008].

With recent advances in machine learning (ML), the development of sophisticated deep learning (DL) models, and the increasing availability of large datasets, data-driven AI approaches are gaining traction, offering promising improvements to traditional forecasting methods. AI-based weather prediction (AIWP) models using transformers [Bi et al., 2023, Chen et al., 2025, 2023, Allen et al., 2025], Fourier neural operators [Bonev et al., 2023, Kurth et al., 2023], and graph neural networks [Keisler, 2022, Lam et al., 2023] have pushed the skill of weather forecasting over what is considered the state of the art for NWP models. Moreover, the recent operational integration of the AI Forecasting System (AIFS), developed by the European Centre for Medium-Range Weather Forecasts (ECMWF), alongside traditional NWP models, marks a significant milestone in the evolution of AIWP [Lang et al., 2024].

Generative approaches have also been introduced, making use of diffusion-based models such as GenCast [Price et al., 2025], which have the ability to directly produce a distribution of potential future weather states. Similarly, Li et al. [2024] presented SEEDS, a Scalable Ensemble Envelope Diffusion Sampler to generate an arbitrarily large ensemble conditioned on as few as one or two forecasts from an operational NWP system. Furthermore, hybrid approaches like NeuralGCM [Kochkov et al., 2024] combine the dynamical core of traditional NWP models with local ML-based parameterizations, achieving performances comparable to operational ensemble forecasts.

Deterministic AIWP models now match or exceed deterministic NWP on standard skill metrics [Rasp et al., 2024], and have been shown to capture and appropriately simulate physical laws in the atmosphere [Baño-Medina et al., 2025a]. Their performance on extreme events, however, is more nuanced. Because they are typically trained with a deterministic objective that rewards variance reduction, they tend to smooth out the tails of the predictive distribution [Brenowitz et al., 2025]. Olivetti and Messori [2024] found that deterministic AIWP models remain competitive with deterministic NWP for extremes, but their relative skill depends strongly on variable, region, and lead time, performing best for temperature, at short lead times, and near the tropics. Zhang et al. [2025] and Zhao et al. [2025] report a similar variable-dependent pattern, with stronger AIWP skill for warm extremes than for wet extremes, and a widening gap relative to NWP at the most extreme tails.

Deterministic skill alone, however, is insufficient for operational use. Because the atmosphere is chaotic, any single forecast will diverge from reality at longer lead times, and stakeholders need not just a best estimate but the range of plausible outcomes and their probabilities [Garg et al., 2022]. This is especially true for extreme events, which are rare by definition and frequently missed by deterministic forecasts at longer horizons. Probabilistic forecasting, through ensembles, as exemplified by the ECMWF Integrated Forecasting System Ensemble (IFS ENS; Molteni et al. [1996]), hereafter referred to as ENS, is therefore widely recognized as the more appropriate framework for high-impact situations, supporting decision-making under uncertainty and enabling authorities to take preventive measures to protect lives and infrastructure [Ponzano et al., 2025].

Earlier work on AIWP was clearly centered on deterministic weather forecasting due to the wide availability of data and benchmarks for such forecasts [Rasp et al., 2024]. However, there has recently been a growing body of work focused on probabilistic AIWP either by introducing post-hoc approaches akin to uncertainty quantification [Bülte et al., 2026], novel perturbation methods similar to those used in traditional NWP ensembles [Mahesh et al., 2025a], by applying Monte Carlo dropout [Garg et al., 2022] or training the model using a probabilistic objective through changes in the loss function, like in AIFS ENS [Lang et al., 2026, Zhong et al., 2025, Alet et al., 2025], or by using generative diffusion AI models to directly produce probabilistic outputs [Price et al., 2025, Couairon et al., 2024]. Recent work suggests that the scarcity and imbalance of extreme events in training datasets can limit model generalization on tail events [Sun et al., 2025].

Uncertainty quantification (UQ) can be done using post-hoc or following approaches based on initial conditions. Post-hoc methods rely on an established methodology to infer a probability distribution from a given deterministic model output. An example of such methods is the EasyUQ method [Walz et al., 2022]. While post-hoc methods are attractive in theory since they are simpler to implement, they make it harder to disentangle sources of uncertainty in the model. Initial condition-based approaches involve making changes to the model input and producing an ensemble of forecasts to infer a probabilistic distribution. While previous studies have explored individual techniques such as Bred Vectors on single architectures [Baño-Medina et al., 2025b, Mahesh et al., 2025a], no prior work has compared how different AIWP architectures respond to a common set of perturbation strategies, nor quantified the relative contributions of model choice and perturbation choice to ensemble skill on extremes.

This paper aims to drive the discussion on the usage of AIWP models, specifically in the context of extreme weather forecasts, by comparing how different AIWP architectures respond to different input perturbation methods. With this aim, we hope to contribute to the following key research questions:

1. How do perturbation-based ensembles built from deterministic AIWP models compare to natively probabilistic systems such as IFS ENS and AIFS ENS in capturing extremes, and how does this gap evolve with lead time and event rarity?
2. Do AIWP ensembles capture different types of meteorological extremes with comparable skill, or are some variables systematically harder than others?
3. To what extent is ensemble skill on extremes governed by the choice of model architecture versus the choice of perturbation method?

By addressing these questions, our study seeks to advance the understanding of both the potential and the limitations of deterministic data-driven models in capturing extreme events through UQ. We assess these questions through a comparison of multiple AIWP architectures under several perturbation strategies, evaluated globally across multiple variables and percentile thresholds, and illustrated in detail on two case studies: the 2022 Pakistan floods and the August 2022 China heatwave. We anticipate that the insights gained in this study will contribute to the practical deployment of data-driven models in operational settings.

2 Data

Two data sources are considered in this study: 1) ERA5 reanalysis data [Hersbach et al., 2020, Carver and Merose, 2023], which serves both as initial conditions for our forecasting models and as reference to evaluate the forecast performance; and 2) two ensemble forecasting systems used as benchmarks: the ECMWF Integrated Forecasting System Ensemble (IFS ENS) [Molteni et al., 1996, ECMWF, 2016], representing a state-of-the-art numerical weather prediction ensemble, and the AIFS ENS [Lang et al., 2026], representing a natively probabilistic AI-based ensemble. IFS ENS forecasts were obtained from the WeatherBench 2 archive [Rasp et al., 2024]. AIFS ENS forecasts were generated by the authors using the publicly released model weights, initialized from ERA5 to ensure a consistent initialization across all AIWP models evaluated in this study. We use a subset of the ERA5 ARCO [Carver and Merose, 2023] variables required for prediction and evaluation, following Bülte et al. [2026], with 6-hourly temporal resolution (see Table 1 for the complete list). To enable a fair comparison between AIWP models trained in the same dataset, all data used in this study are downscaled to a common 1°× 1° spatial resolution, using bilinear interpolation [Geneva and Foster, 2024]. The evaluation is conducted on: Geopotential at 500 hPa ($Z500$), Temperature at 850 hPa ($T850$), 2 m Temperature ($T2M$), U and V components of wind at 10 m ($U10M$, $V10M$), Total Precipitation (6h accumulation, TP_{6h}), and additionally 24h aggregations for total precipitation (TP_{24h}) and maximum for all other evaluation variables ($T2M_{24h}$, $T850_{24h}$, $Z500_{24h}$, $U10M_{24h}$, $V10M_{24h}$).

3 Methodology

Figure 1 illustrates the methodology followed in this study. First, starting from ERA5 reanalysis data as initial conditions, we apply several perturbation methods, each of them generating $M = 50$ different initial states. Then, these states are provided to diverse 6-hourly resolution AIWP models, resulting in $M = 50$ different predictions, which constitute ensemble forecasting members for up to a 10-day lead time. The ensembles for all perturbation methods and AIWP models, along with ENS and AIFS ENS, are finally evaluated 6-hourly and at daily resolution against the reference ERA5 to assess their forecasting capabilities.

For data handling and running predictions with various models, we use the NVIDIA-developed Earth2Studio framework [Geneva and Foster, 2024]. This modular and Python-based framework enables running large-scale inference, with multiple models and perturbation methods. In addition, for the evaluation of the forecasts, we make use of the WeatherBench framework [Rasp et al., 2024] and the probabilistic metrics extension to WeatherBench introduced in Zhao et al. [2025].

3.1 Models

We select three AIWP models for the purpose of this study that fulfill the following conditions:

Table 1: ERA5 variables used for forecasting with AIWP models.

Variable	Description	Unit	Pressure levels (hPa)
z (levels)[a]	Geopotential	$m^2\,s^{-2}$	1000, 925, 850, 700, 600, 500, 400, 300, 250, 200, 150, 100, 50 (13 levels)
q	Specific humidity	$kg\,kg^{-1}$	
r[b]	Relative humidity	%	
t[a]	Temperature	K	
u[a]	U component of wind	$m\,s^{-1}$	
v[a]	V component of wind	$m\,s^{-1}$	
w	Vertical velocity	$Pa\,s^{-1}$	
msl[a]	Mean sea level pressure	Pa	
u10m[a]	10m U component of wind	$m\,s^{-1}$	
v10m[a]	10m V component of wind	$m\,s^{-1}$	
u100m	100m U component of wind	$m\,s^{-1}$	
v100m	100m V component of wind	$m\,s^{-1}$	
t2m[a]	2m Temperature	K	
d2m	2m Dewpoint temperature	K	
stl1	Soil temperature level 1	K	
stl2	Soil temperature level 2	K	
sp	Surface pressure	Pa	
tcwv	Total column water vapour	$kg\,m^{-2}$	
tcw	Total column water	$kg\,m^{-2}$	
swvl1	Volumetric soil water layer 1	$m^3\,m^{-3}$	
swvl2	Volumetric soil water layer 2	$m^3\,m^{-3}$	
tp06[c]	Total precipitation (6h accumulation)	m	
z (surface)	Geopotential at surface	$m^2\,s^{-2}$	
lsm	Land–sea mask	(0–1)	
sdor	Standard deviation of orography	m	
slor	Slope of sub gridscale orography	m	

[a] This variable is perturbed when using perturbation methods. [b] This variable is derived from pressure, **q**, and **t**. [c] This variable is accumulated in 6-hourly intervals from the original 1-hourly data.

1. Perform at/close to the state of the art for deterministic metrics and medium-range weather forecasting;
2. Have both model code implementation and weights available in an open source way;
3. Produce forecasts for total precipitation and surface temperature, given that we want to understand the different models' prediction behavior concerning extreme precipitation and heatwaves, respectively.

Based on these criteria and our prior literature review, as well as taking into account diversity in model architecture type, this study uses three different architectures: FuXi [Chen et al., 2023], GraphCast [Lam et al., 2023], and SFNO (Spherical Fourier Neural Operator) [Bonev et al., 2023]. For comparison, we also evaluated AIFS ENS [Lang et al., 2026], a probabilistically trained AIWP example model. Below, we summarize these models to highlight their architectures and forecasting approaches, referring the reader to their original publications for further details.

FuXi is a hybrid architecture combining a cascading U-Transformer with Swin Transformer V2 blocks to deliver high-accuracy forecasts across multiple time scales. FuXi uses three distinct sets of model weights to predict the weather between 0-5 days (FuXi short), 5-10 days (FuXi medium), and over 10 days (FuXi long) [Chen et al., 2023]. As input, it takes two timesteps to predict the subsequent meteorological state ($X^{t-1}, X^t \to X^{t+1}$).

GraphCast builds on the pioneering work of Keisler [2022] using a GNN based on an encoder, processor, and decoder architectures that include simultaneous multi-mesh message passing at different scales [Lam et al., 2023]. It relies on two timesteps as input to predict the following timestep ($X^{t-1}, X^t \to X^{t+1}$), along with multiple forcing fields like time of day, day of year, and the total top of atmosphere incident solar radiation, among others. GraphCast has been trained on ERA5 data at 1° resolution and fine-tuned for IFS initial

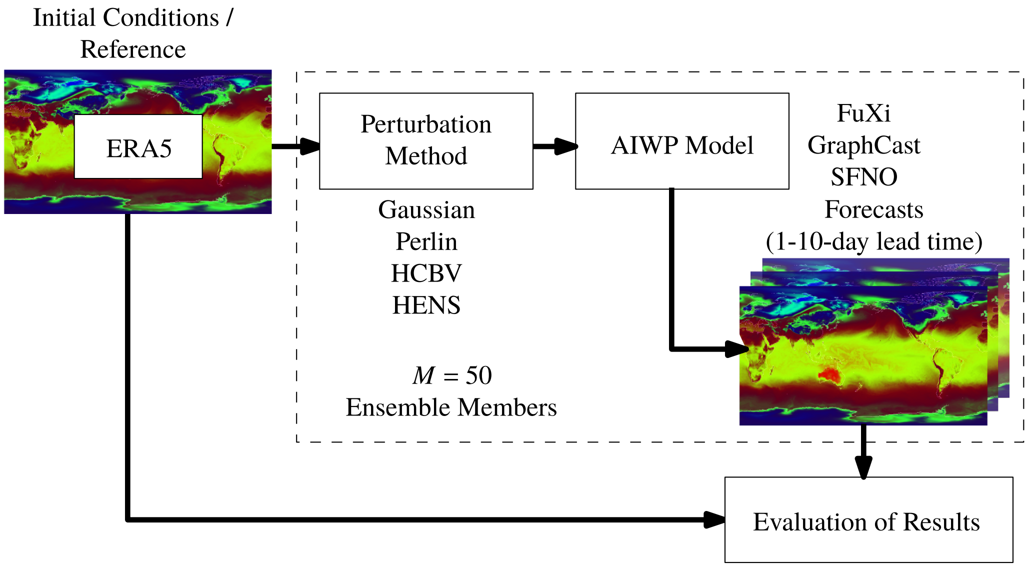

Figure 1: Overview of the study methodology, including the initial condition perturbation, forecasting with AIWP models, and evaluation of results stages.

conditions at IFS-like resolutions (0.25°). As stated in the previous section, we decided to use the 1° model for GraphCast in order to enable a fair comparison with all other models, which have not been finetuned on IFS initial conditions.

SFNO is a Spherical Fourier Neural Operator-based architecture that is designed to leverage the properties of the Fourier operators on spherical geometries, modeling long-range dependencies in spatio-temporal data via convolutions [Bonev et al., 2023]. This model does not predict total precipitation directly, but comes with a trained diagnostic model [NVIDIA NGC Catalog, 2025] that takes in the output of the SFNO model and produces a total precipitation output ($X^t \rightarrow X^{t+1}, \omega(X^{t+1})$, where ω is the diagnostic total precipitation model).

While this study focuses on three deterministic, state-of-the-art AIWP models, we acknowledge that other architectures could also have been included. In particular, models such as AIFS [Lang et al., 2024] and NeuralGCM [Kochkov et al., 2024] represent important advances and improvements in forecast skill. Nevertheless, we believe that the selected set provides a representative cross-section of current state-of-the-art approaches across transformer-, GNN-, and operator-based architectures, allowing for a comparison of how leading AIWP models respond to input perturbations.

3.2 Perturbation Methods

We select four perturbation methods for this study, spanning a range of complexity from simple stochastic baselines to flow-dependent approaches. The first two methods, Gaussian noise and Perlin noise, generate spatially correlated perturbations that are independent of the atmospheric state, differing in how spatial correlation is introduced: through a prescribed covariance function (Gaussian) or through procedural interpolation on a lattice (Perlin). The remaining two methods, the Hemispheric Centered Bred Vector (HCBV) and Huge Ensembles (HENS), are both based on bred vectors, a class of flow-dependent, nonlinear perturbations designed to capture the fastest-growing modes of the atmospheric flow [Feng et al., 2018, Toth and Kalnay, 1997]. HCBV follows the formulation of Baño-Medina et al. [2025b], while HENS incorporates additional model-specific amplitude scaling as introduced by Mahesh et al. [2025a]. This selection allows us to assess both whether spatial correlation configuration in the perturbations matters (Gaussian vs. Perlin) and whether flow-dependent structure provides additional skill beyond appropriately scaled stochastic noise (Gaussian/Perlin vs. HCBV/HENS).

For all perturbation methods, only the input variables common to all three AIWP models are perturbed (see Table 1). Each method relies on a dimensionless scaling factor s that controls the overall perturbation magnitude. To select an appropriate value, we performed a sensitivity analysis over s for each perturbation method (see Appendix E). Based on this analysis, we set $s = 0.35$ for Gaussian and HENS perturbations. These scaling factors are consistent with the scaling used in Mahesh et al. [2025a]. For HCBV, we found the optimal scale factor as one order of magnitude smaller ($s = 0.035$), while for Perlin noise the optimal factor is one-quarter of the 0.35 ($s = 0.0875$). Example input perturbations for each of the methods described are available in Appendix D.

3.2.1 Gaussian Noise

The Gaussian perturbation method generates ensemble members by adding a realization of a Gaussian random field on the sphere to the initial conditions. This field is sampled spectrally using spherical harmonics, with a Matern covariance function that controls the spatial correlation structure. The perturbed initial condition at each grid point is given by:

$$y_{\text{pert}} = y + s \cdot \sigma_{\text{clim}} \cdot \xi \tag{1}$$

where y is the unperturbed initial condition, s is a dimensionless scale factor, σ_{clim} is the climatological standard deviation of the variable (computed from the ERA5 1990–2020 climatology), and ξ is a mean-zero, unit-variance spherical Gaussian random field evaluated at each latitude and longitude. The Matern covariance uses the correlation length scale of 3 and smoothness of 4 set by the default configuration of the `SphericalGaussian` implementation in Earth2Studio [Geneva and Foster, 2024].

3.2.2 Perlin Noise

Perlin noise is a procedural noise function that generates spatially correlated random fields by interpolating between pseudo-random gradient vectors on a regular lattice, following the configuration in Bi et al. [2023]. Unlike the Gaussian method, which derives its correlation structure from a prescribed covariance function, Perlin noise produces spatial coherence through local interpolation. The perturbed initial condition is given by:

$$y_{\text{pert}} = y + s \cdot \sigma_{\text{clim}} \cdot P \tag{2}$$

where $P \in [-1, 1]$ is the Perlin noise value at each latitude and longitude, and all other terms are as defined above. Each ensemble member uses an independent random seed, producing a unique perturbation field.

3.2.3 Hemispheric Centered Bred Vector (HCBV)

The bred vector algorithm, introduced by Toth and Kalnay [1993], estimates the fastest-growing perturbation modes by exploiting the fact that initial conditions produced by data assimilation accumulate growing errors. Starting from a small seed perturbation, the method uses the forecast model itself to identify the directions in the state space along which errors amplify fastest.

A seed perturbation Δz_{500} is generated by adding a correlated spherical Gaussian noise field (with σ_{clim} amplitude) to the 500 hPa geopotential variable. The AIWP model is then run from both the perturbed and unperturbed initial conditions to produce forecasts f_p and f_u, respectively. The raw bred vector is computed as the difference between these two forecasts, normalized separately over each hemisphere and scaled:

$$\Delta f = s \cdot \sigma_{\text{clim}} \cdot \frac{f_p - f_u}{h(f_p - f_u)} \tag{3}$$

where s is the dimensionless scale factor, σ_{clim} is the climatological standard deviation of the variable (computed from the ERA5 1990–2020 climatology), and $h(\cdot)$ denotes the hemispheric norm, computed separately for the North and South regions (defined as $|\text{latitude}| > 70°$) and interpolated in the tropics, following Mahesh et al. [2025a].

To better capture the dominant growing modes, this breeding cycle is repeated $d = 3$ times: at each cycle, the bred vector from the previous iteration is used as the new seed perturbation, and the model is run forward for one autoregressive step (6 h) to recompute the normalized difference. The three recursive cycles span a total breeding period of 18 h, yielding the final perturbation Δf_d. This integration depth follows Baño-Medina et al. [2025b] and is intended to allow the perturbation to project more strongly onto the dynamically unstable modes of the atmosphere.

The perturbation is additionally centered: for each pair of ensemble members, the bred vector Δf_d is alternately added to and subtracted from the initial conditions, ensuring that the ensemble mean remains unbiased with respect to the unperturbed analysis. The method is illustrated in Fig. 2.

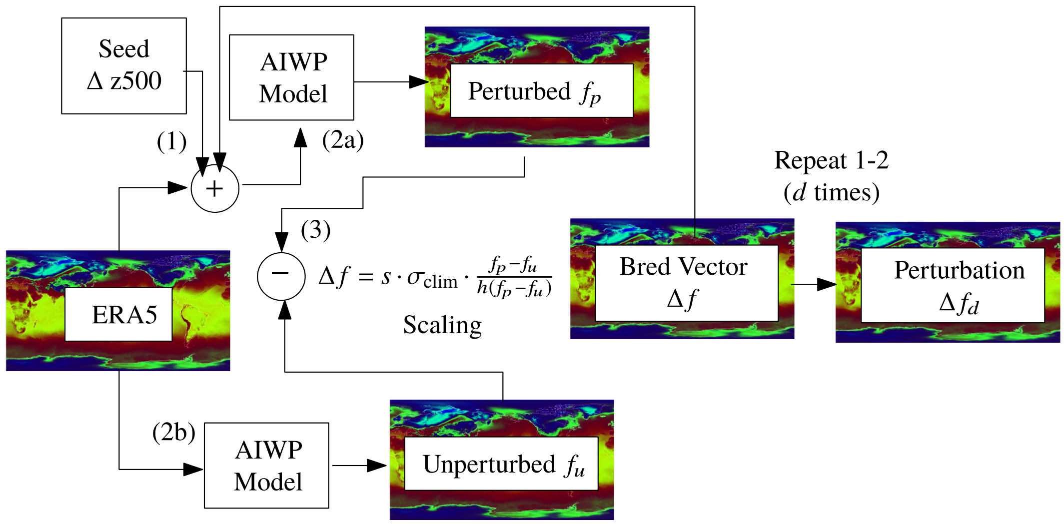

Figure 2: Hemispheric Centered Bred Vector (HCBV) perturbation method. A seed perturbation Δz_{500}, generated as a correlated spherical Gaussian noise field, is added to the 500 hPa geopotential variable. The AIWP model produces perturbed (f_p) and unperturbed (f_u) forecasts, whose difference is normalized separately for the North and South regions ($|\text{latitude}| > 70°$) with interpolation in the tropics, denoted by $h(\cdot)$. The result is scaled by the dimensionless factor s and the climatological standard deviation σ_{clim}. This breeding cycle is repeated $d = 3$ times, each corresponding to one 6 h autoregressive model step (18 h total), yielding the final perturbation Δf_d. The perturbation is additionally centered: Δf_d is alternately added and subtracted from the initial conditions. Diagram adapted from Baño-Medina et al. [2025b].

3.2.4 Huge Ensembles (HENS)

Huge Ensembles (HENS) is an adapted version of the HCBV method, introduced by Mahesh et al. [2025a], that incorporates model-specific amplitude tuning to better match the expected forecast error growth of each AIWP architecture. The key modification is in the scaling of the perturbation: HCBV uses a climatological scaling, HENS replaces this with a model dependent amplitude:

$$\Delta f = s \cdot \text{RMSE}_{48h} \cdot \frac{f_p - f_u}{h(f_p - f_u)} \tag{4}$$

where RMSE_{48h} is the global root mean squared error of the deterministic AIWP model at a 48 h lead time for the perturbed variable, and s is the dimensionless scale factor. This scales the perturbation to match with the actual forecast error magnitude of each model, so that models with larger intrinsic errors receive proportionally larger perturbations. The same model-specific scaling is also applied to the seed perturbation Δz_{500}.

It should be noted that the original HENS implementation [Mahesh et al., 2025a] also employs an ensemble of model checkpoints to produce the final ensemble prediction, combining both initial-condition and model-induced diversity. In this study, we isolate the HENS initial-condition perturbation method only, in order to enable a direct comparison between perturbation strategies.

3.3 Evaluation Metrics

We select three complementary evaluation metrics to assess both deterministic and probabilistic forecast skill.

The **Root Mean Squared Error (RMSE)** measures the average magnitude of errors between the ensemble mean forecast or deterministic forecast and the reference values, penalizing larger errors more heavily. Lower RMSE indicates better deterministic accuracy. An RMSE of zero corresponds to a perfect forecast.

The **Continuous Ranked Probability Score (CRPS)** evaluates the full predictive distribution by measuring the integrated squared difference between the forecast cumulative distribution function and the reference. It generalizes the mean absolute error to probabilistic forecasts, simultaneously rewarding both narrow predictive distributions and statistical consistency with the reference. Lower CRPS indicates better probabilistic skill, with zero representing a perfect probabilistic forecast.

The **Receiver Operating Characteristic Skill Score (ROCSS)** quantifies the ability of a forecast to discriminate between the occurrence and non-occurrence of a binary event, relative to random chance. It is derived from the Area Under the ROC Curve (AUC), which measures the trade-off between hit rate and false alarm rate across all probability thresholds. A ROCSS of 1 indicates perfect discrimination, 0 indicates no skill, and negative values indicate performance worse than random chance. This metric becomes relevant for extremes through the choice of an event threshold τ. In this study, we define events using climatological percentiles of the ERA5 1990–2020 distribution (primarily $\tau = P_{99}$, with additional analysis at the 90th, 95th, 98th, and 99.5th percentiles), so that the ROCSS specifically measures discrimination skill for rare, high-impact events. Throughout this study we made use of daily aggregates (sum for precipitation, maxima for all other variables) for computing ROCSS following Zhao et al. [2025].

To quantify the relative importance of model architecture versus perturbation method in determining ensemble skill, we apply a two-way analysis of variance (ANOVA) to each forecast metric and report the variance fraction η^2 attributable to model architecture, perturbation method, and their interaction. Variance-partitioning approaches of this kind have a precedent in the climate-projection literature, where ANOVA frameworks are used to decompose total ensemble variance into contributions from modeling factors and their interactions [Yip et al., 2011].

Formal definitions of all three metrics and the ANOVA approach are provided in Appendix A.

4 Case studies

Two case studies are considered: the Pakistan Floods and the China Heatwave in August 2022. They are illustrated in Figure 3. For our purposes, only data within the geographical bounds displayed in the figure and corresponding to August 2022 is used. For both the Pakistan floods and the China heatwave, we use the climatological 99th-percentile threshold to isolate and define the extremes, consistent with precipitation analyses in Singh et al. [2024] and heatwave characterizations in Sun et al. [2021].

4.1 Summer 2022 Pakistan Floods

In the summer of 2022, Pakistan experienced catastrophic flooding that submerged one-third of the country, resulting in over 1000 deaths, displacing 30 million people, and causing over 30 billion USD in damages [Arshad et al., 2025]. The rainfall observed in this period was four standard deviations above the climatological mean and twice the amount of a previous flooding event in 2010 [Hong et al., 2023]. Recent studies have linked this extreme event with concurrent heatwaves [Hong et al., 2023], which favored snow melt, subsequently followed by extreme precipitation [Shehzad, 2023]. During that time, parts of the territory experienced daily precipitation amounts above 100 mm, which, according to Ikram et al. [2016], is considered extreme in Pakistan. Large parts of Pakistan with daily precipitation exceeding 100 mm during August 2022 can be observed in Figure 3 corresponding to the 99th percentile of the 1990-2020 climatology of daily precipitation.

4.2 August 2022 China Heatwave

During August 2022, southern China was hit with a record-breaking heatwave, with over 360M people experiencing temperatures above 40°C [Gong et al., 2024]. This heatwave represented the greatest anomaly since 1979 and caused a severe impact, including massive droughts [Zhou et al., 2023, Sun et al., 2023]. As can be seen in Figure 3, the region presented close to 10°C exceedance with respect to the 99th-percentile of the ERA5 1990-2020 climatology for the daily maximum temperature on the majority of days in August 2022, with an even greater exceedance in the Sichuan Basin.

5 Results

We begin this section by evaluating the performance of the three selected deterministic AIWP models in predicting extreme events under input perturbations. The resulting ensembles exhibited varying skill depending on the predicted variable, model architecture, perturbation strategy, and forecast lead time. In the following figures, we analyze the ROCSS for 90th to 99.5th percentile thresholds. Then, for each case

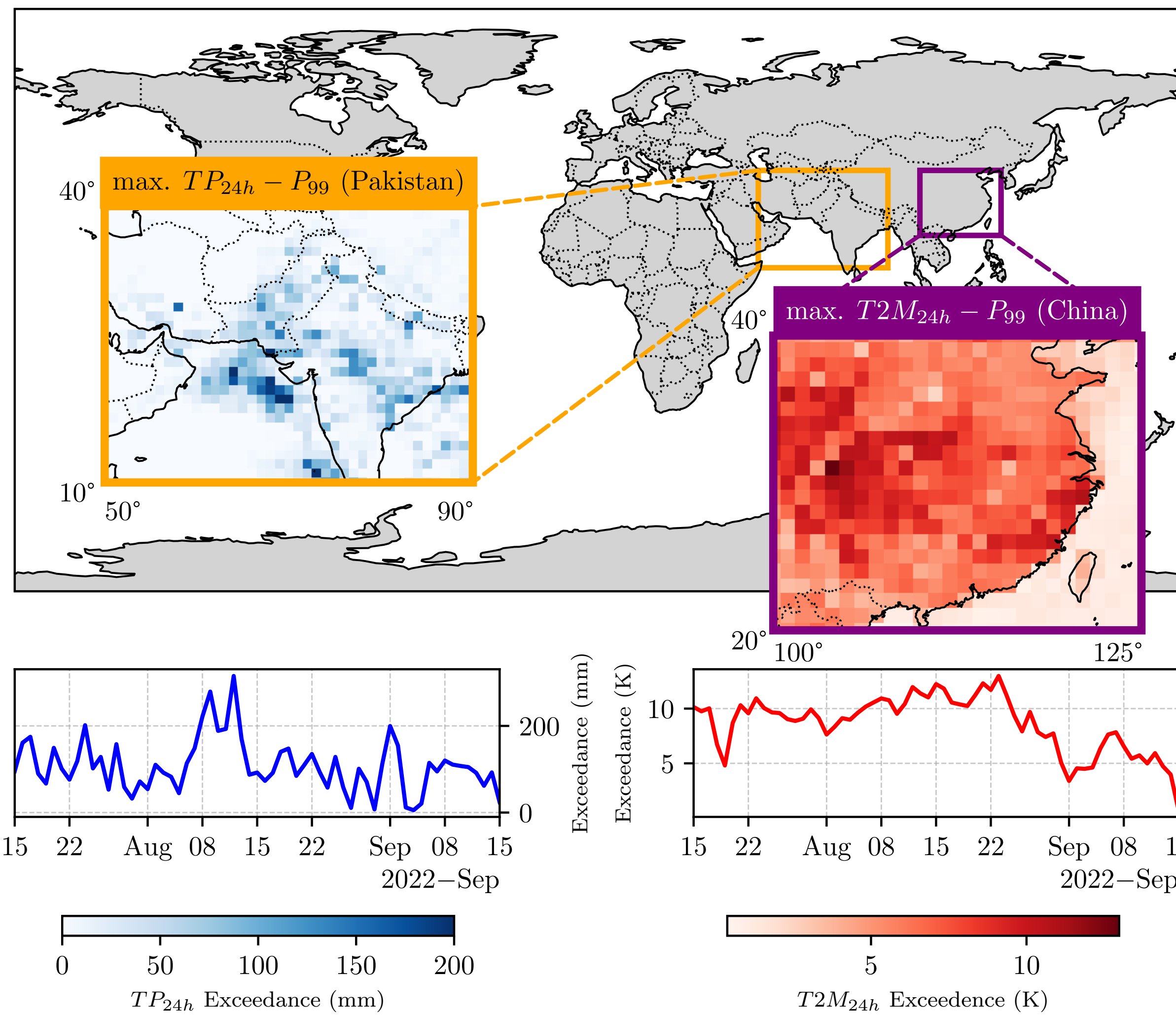

Figure 3: Analysis of the August 2022 Pakistan extreme precipitation and China heatwave, based on ERA5. The values displayed (in the spatial and temporal dimensions) correspond to the exceedance of the 99th percentile of the ERA5 1990-2020 climatology for daily total precipitation and daily maximum temperature. The geographical bounds of each case study are also shown. A maximum of 200 mm of precipitation exceedance can be appreciated in Pakistan, while a sustained 10 K exceedance over August is observed in China, further reinforcing the extreme nature of both events.

study, we analyze the ROCSS skill and inspect the spatial spread characteristics. Finally, we evaluate the relative contribution of model and perturbation choice in the forecast skill. In addition, we explore the links between the performance in these case studies and global evaluation metrics like CRPS and RMSE.

Globally, in Fig. 4, we can conclude that discrimination skill decreases as percentile increases across models and variables (i.e., forecast skill is reduced in more extreme events). Overall, forecast skill for total precipitation is the lowest, followed by 10m wind vectors. Geopotential shows the best discrimination skill. Looking at the model, we can see stark performance differences across models for precipitation. Performance decreases with lead time, with FuXi displaying the highest spread across lead times. ENS and AIFS ENS outperform on average for discrimination skill. Analyzing perturbations, we observe that the shape of the spread across perturbation methods is very similar, even in precipitation, where the most pronounced model differences are found. This indicates a comparatively small effect of perturbation choice on forecast skill.

Figure 5 displays our case study evaluation, with the ROCSS at the 99th-percentile for daily precipitation over Pakistan and daily maximum temperature over China. Overall, ensemble skill decreases with lead time,

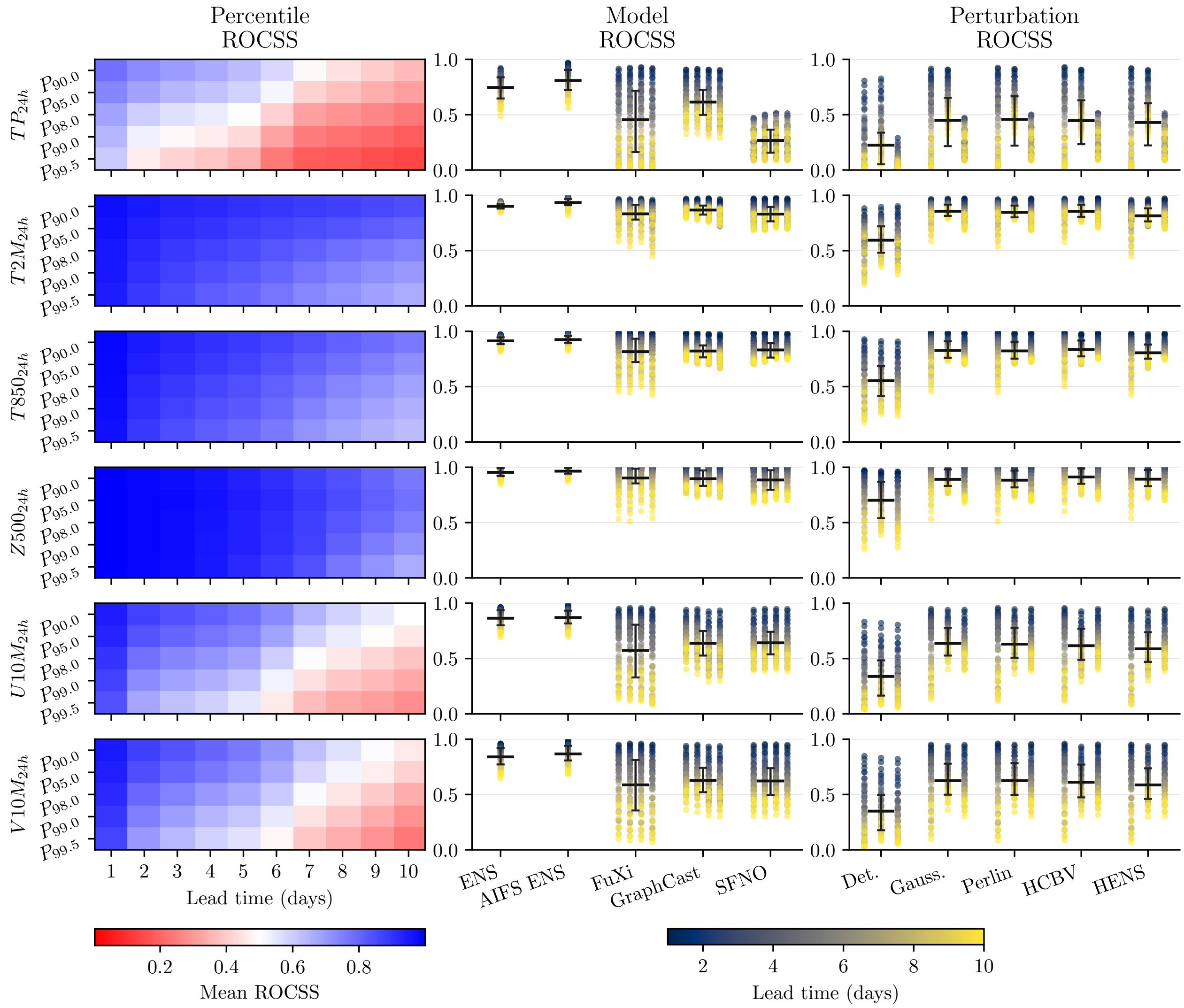

Figure 4: Global ROCSS decomposed by percentile threshold, model architecture, and perturbation method across August 2022. In the left column, we display mean ROCSS for models and perturbations as a function of lead time and percentile threshold; blue indicates high skill, red low skill. In the middle and right columns, ROCSS per model and ROCSS per perturbation, with AIFS ENS and ENS as benchmarks, and deterministic forecasts. Each dot in this plot represents the metric for a given lead time, percentile, model, and perturbation, with each column grouping dots per perturbation (Gaussian, Perlin, HCBV, and HENS) or model (FuXi, GraphCast, SFNO). The similar spread of ROCSS in the right column indicates that model architecture is the dominant factor governing discrimination skill, while the choice of perturbation method has a comparatively small effect.

with the exception of SFNO for precipitation, reflecting the limited predictability horizon of these AIWP and NWP systems. For both ENS and AIFS ENS, we notice a slight increase in performance for precipitation from 1 to 2 lead times, most likely because ENS is initialized with IFS initial conditions and AIFS ENS is also fine-tuned with that data, which causes a small dip in performance in the first forecast step when compared with models trained with ERA5. However, both ENS and AIFS ENS showcase the best metrics across variables and lead times, with AIFS ENS showing significantly higher ROCSS than ENS up to a 5-day lead time for precipitation and 6-day lead time for temperature, demonstrating the benefits of a probabilistically trained model. Among the perturbation methods, Gaussian, Perlin, and HCBV achieved similarly high and the highest probabilistic skill on these extreme events, and all perturbation methods improved upon the model's deterministic skill.

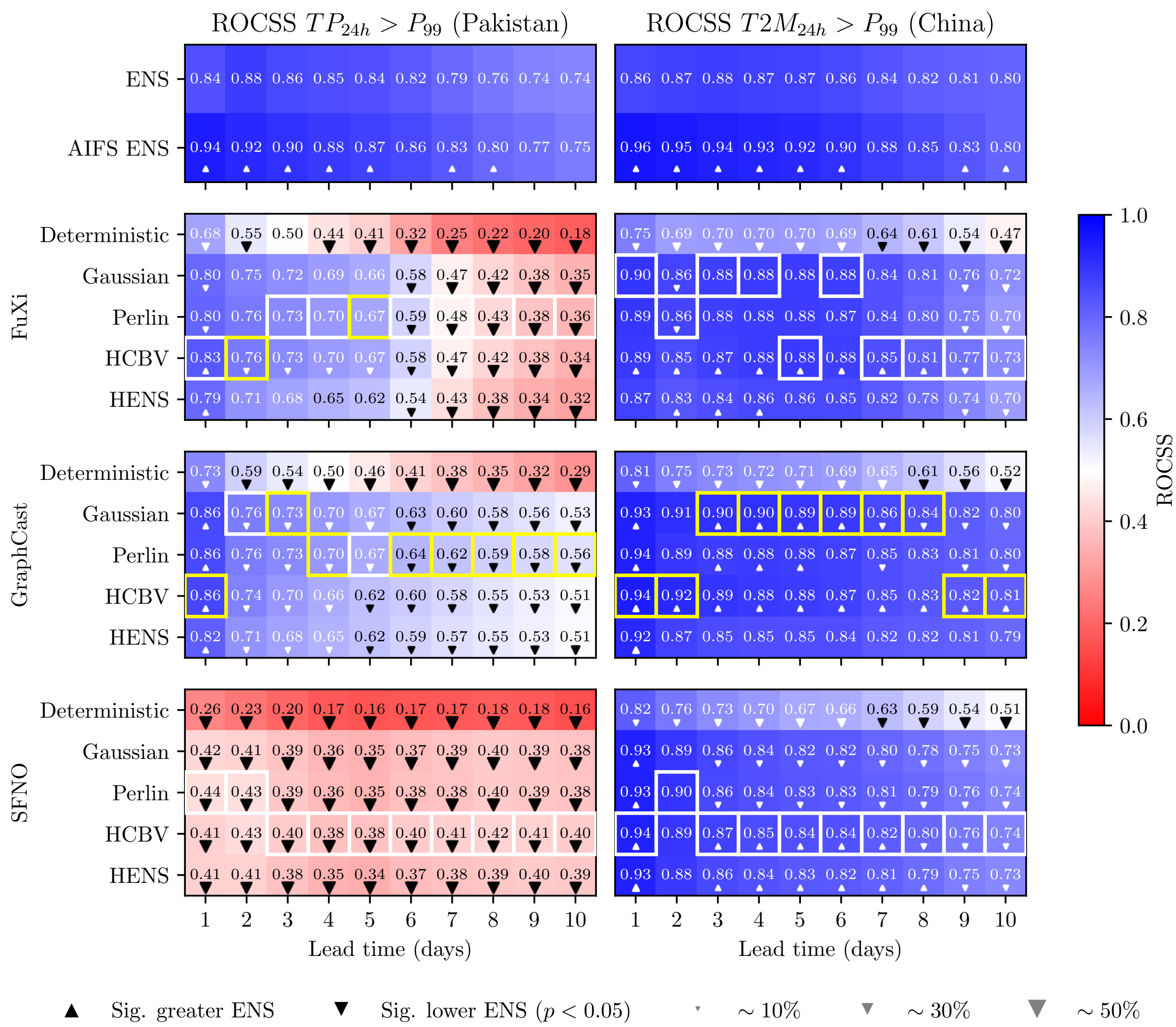

Figure 5: ROCSS at the 99th percentile for daily precipitation (Pakistan, left) and maximum daily temperature (China, right) in August 2022. ENS and AIFS ENS benchmarks are shown at the top, followed by each AIWP model with its deterministic forecast and four perturbation methods. Triangles indicate statistically significant differences relative to ENS ($p < 0.05$): upward for higher, downward for lower, with size proportional to the magnitude of the difference. Yellow boxes represent the highest metric overall, and white boxes highlight the highest metric for each model. All perturbation methods improve upon the deterministic baseline, but differences between model architectures are substantially larger than differences between perturbation strategies.

For very high precipitation (i.e., 99th percentile) in Pakistan, GraphCast–HCBV (and Gaussian and Perlin) reaches ROCSS ≈ 0.86 at lead time of 1 day, the highest ROCSS for this variable. At longer lead times, GraphCast–Perlin has the best discrimination skill, keeping ROCSS above 0.6 until 7 days lead time. In contrast, SFNO ensembles exhibit weak discrimination skill for extreme precipitation (ROCSS < 0.44), indicating very limited performance for precipitation extremes. Interestingly, the performance for SFNO overall decreases around a 4-5-day lead time and recovers around the 8-day lead time for HCBV perturbations. SFNO uses a separate diagnostic model to produce precipitation forecasts. This diagnostic model was trained using ERA5 data [NVIDIA NGC Catalog, 2025], not SFNO forecasts, which could potentially explain the poor performance of SFNO for precipitation overall.

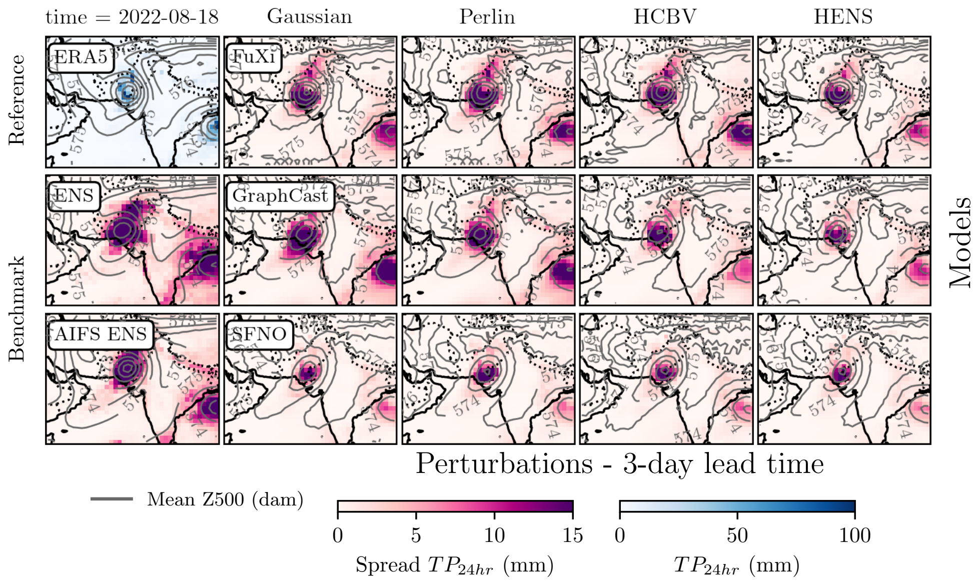

Figure 6: Daily accumulated precipitation spread for the different ensemble models (ENS, AIFS ENS, AIWPs) over Pakistan on 18th August 2022 for a 3-day lead time forecast. ERA5 spatial distribution for daily precipitation is shown in the top left corner. Daily average geopotential height at 500 hPa is also displayed as contour lines. FuXi and GraphCast with Gaussian/Perlin noise most closely match the ENS spread. Within each model, the perturbations look very similar.

For temperature extremes (i.e., 99th percentile) in China, GraphCast–Gaussian performs best, attaining ROCSS ≈ 0.94 at a lead time of 1 day, nearly matching AIFS ENS (0.96), and being significantly higher than ENS up to a 6-day lead time. FuXi–HENS shows the worst performance, and it is the only model that does not show significantly higher ROCSS than ENS at any lead time. For precipitation, Perlin and Gaussian perturbations support GraphCast ROCSS for longer lead times than FuXi, which very precipitously decreases performance across lead times, from 0.83 to 0.32. In addition, for all models, ROCSS for temperature is comparatively much more stable across lead times. This pattern suggests that AIWP models are more effective at representing smoother, large-scale thermodynamic fields than precipitation fields, and perturbations alone are not able to overcome that for discrimination skill, although they do seem to support GraphCast's performance more than FuXi or SFNO.

The spatial ensemble spread at 3-day lead time in Pakistan (Fig. 6) and China (Fig. 9) highlights how model selection plays the biggest role in the ensemble spread. In Fig. 6, we can observe that ENS and AIFS ENS capture the heavy precipitation core and associated uncertainty. The perturbed AIWP ensembles show similar spatial spread composition across perturbations, although only FuXi and GraphCast with Gaussian and Perlin seem to be able to reproduce the uncertainty over Pakistan and in eastern India. The perturbations are however slightly underdispersive when compared with the benchmark. Similar patterns are observed for temperature.

We further assess global performance to determine whether the patterns observed in these case studies extend across all regions and non-extreme settings. Globally averaged metrics reinforce the regional findings (see Appendix B). All perturbations yield similar probabilistic metrics for each model, although Gaussian and Perlin have a slight edge (see Fig. 10). Probabilistic skill is consistently higher for temperature than for precipitation across all models.

In Fig. 7, we investigate the relative role of the perturbation and model choice from a statistical perspective, using analysis of variance. For CRPS and RMSE, the model architecture is overwhelmingly the leading factor and the corresponding spatial maps are almost uniformly model-dominated. Perturbation methods contribute only a thin marginal share (under 10%), and the interaction term is similarly small. The picture changes substantially for the ensemble spread, where the perturbation method explains a large fraction of the variance at short lead times, between 30–50% for $T2M$, $T850$, TP_{6h}, $U10M$ and $V10M$ at 1 day lead time and then decays, so that by day 10, model choice again dominates. Geopotential at 500 hPa ($Z500$) is a

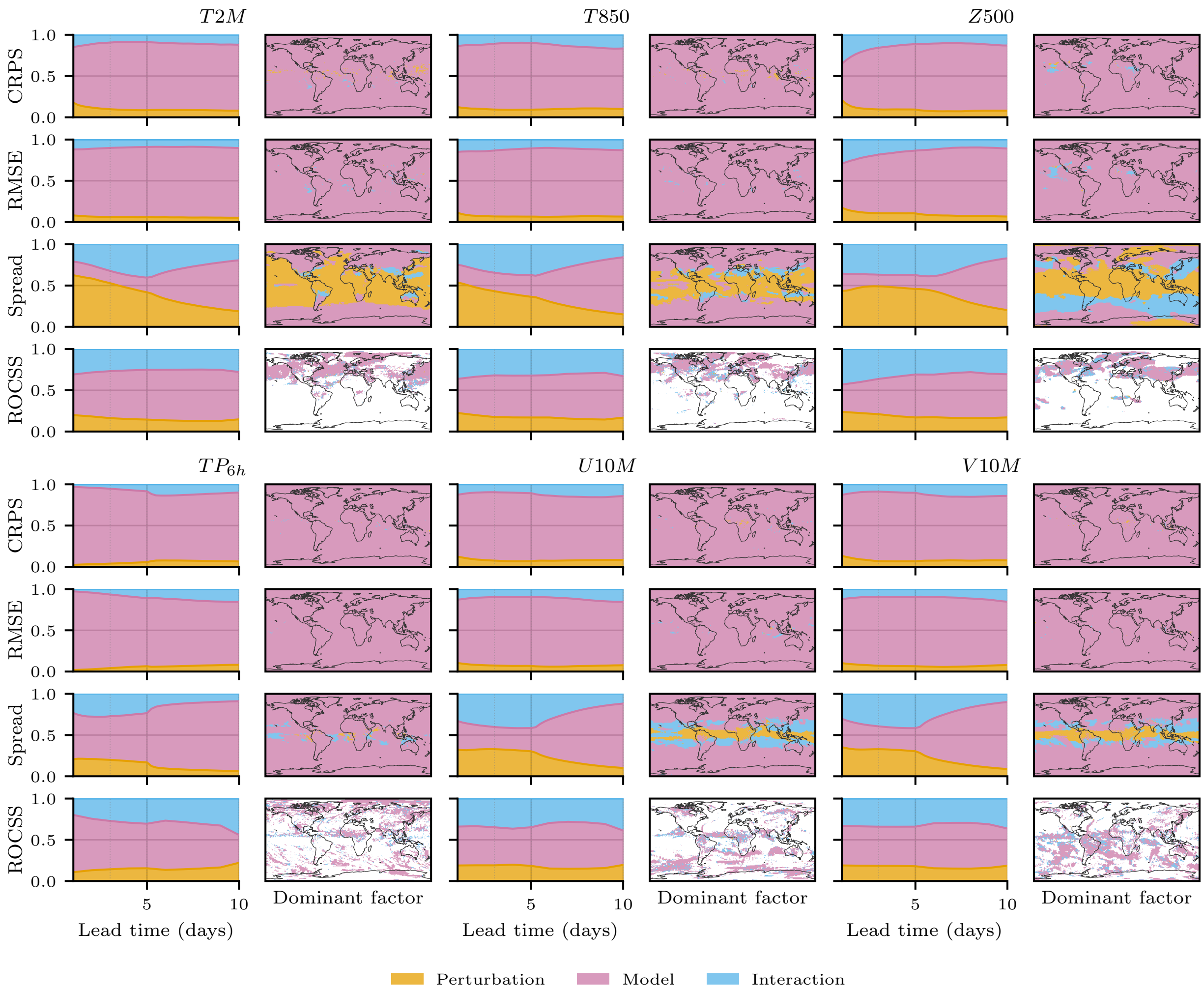

Figure 7: Variance decomposition of forecast performance into contributions from perturbation method, model, and their interaction, using η^2 from a two-way ANOVA (see Appendix A). For each variable, results are shown for CRPS, RMSE, ensemble spread and ROCSS with left panels representing area-weighted global mean η^2 as a function of lead time and the right panel a spatial map of the dominant factor (perturbation, model, or interaction) at each grid point, determined by the largest η^2 component, averaged across lead times. For ROCSS we used the 99th percentile, and the daily aggregates for each variable. Model architecture (pink) dominates CRPS and RMSE across nearly all variables and lead times, while the perturbation method (yellow) plays a larger role for ensemble spread, particularly at short lead times and for geopotential ($Z500$).

partial exception: its spread is governed largely by the model and perturbation interaction rather than by either factor alone, with the interaction component visible across much of the global map and particularly over the oceans. For ROCSS, model architecture remains the leading factor across most variables (over 60%), but the interaction term becomes more prominent at longer lead times, and the spatial maps are sparser, reflecting the grid points where the metric is undefined for rare events.

The choice of perturbation method exerts its strongest influence where expected: on the spread of the ensemble at short lead times, when initial-condition uncertainty has not yet been overwhelmed by the model's error growth, but it has comparatively little leverage on the accuracy of the ensemble mean (RMSE) or the overall probabilistic skill (CRPS), both of which governed almost entirely by the model choice. The case of $Z500$ spread, where the interaction term dominates, suggests that some perturbation–model combinations behave non-additively, which can be explained by the fact that HCBV and HENS use the forecast model itself to generate the perturbed states, and $Z500$ is our seeding variable. The decay of the perturbation share with lead time is also consistent with a growing role of model-driven error at longer lead times.

For both case studies and global evaluations, the results reveal a consistent pattern: any of the perturbations studied seem to improve the probabilistic skill among deterministic AIWP models, with simpler methods (Gaussian and Perlin) appearing to have a slight edge. Overall, ensemble skill (ROCSS, CRPS) declines with forecast lead time, reflecting the inherent limits of predictability and the growing influence of model-driven error. AIWP models perform substantially better for temperature extremes (coherent, large-scale events) than for precipitation extremes, which depend on small-scale processes poorly represented in their training data (i.e., ERA5). Globally, CRPS and RMSE metrics confirm that perturbation-based ensembles can emulate aspects of probabilistic behavior but still lack the spatial variability and multiscale energy structure of NWP ensembles or native probabilistic models, particularly for precipitation. Overall, while these perturbations move AIWP deterministic models toward meaningful ensemble forecasting, performance mismatches persist.

6 Discussion

Our results are consistent with recent evaluations of deterministic AIWP models [Olivetti and Messori, 2024, Zhang et al., 2025, Brenowitz et al., 2025], which find that data-driven systems outperform traditional NWP on deterministic metrics but show weaker probabilistic skill. As in Mahesh et al. [2025a,b], we confirm that input perturbations can generate meaningful spread from AIWP architectures. However, the comparable performance of HCBV and HENS to Gaussian and Perlin noise contrasts with Bülte et al. [2026], who reported Random Field Perturbations outperforming Gaussian noise; this discrepancy likely reflects their use of a smaller amplification factor and spatially incoherent noise. Our findings indicate that when perturbations are spatially coherent and appropriately scaled, their specific structure matters less than their amplitude. GraphCast shows an advantage for both precipitation and surface temperature, suggesting benefits from graph-based atmospheric representations.

Ensemble power spectra (Appendix C) show that AIWP ensembles underestimate precipitation spread at all scales relative to ENS, while 2m temperature spectra match more closely—particularly for FuXi–Perlin (Fig. 13). Deterministic AIWP spectra track ERA5 well for temperature, but drop in power at the mesoscale, and SFNO is underpowered across all scales for precipitation. This is consistent with Rodwell et al. [2025].

Two caveats concern our choice of reference and baseline. First, ERA5 has known limitations in capturing precipitation variability in the tropics and regions with sparse observations [Lavers et al., 2022], and the 2022 Pakistan event falls within both categories. Absolute ROCSS and CRPS values may therefore be affected, though the relative ranking of models and perturbations should be more robust since all configurations are evaluated against the same reference. Gauge-based or satellite-derived products (e.g., IMERG) or regional reanalyses with denser observational input would provide a more faithful evaluation [Gupta et al., 2025]. Second, ECMWF's Open Data now provides high-quality ensemble initial conditions that would likely improve performance and represent a stronger operational baseline. We do not use them here because our focus is on self-contained perturbation strategies applicable when operational ensemble analyses are unavailable, such as in historical reforecasts; a direct comparison remains for future work.

The decline in ensemble skill with lead time reflects both initial-condition sensitivity and model-induced error, which our perturbation-based ensembles do not cleanly isolate: Gaussian and Perlin noise only crudely approximate the true initial-condition distribution, while HCBV and HENS use the forecast model itself to generate perturbed states, injecting model-dependent structure. Empirically, the η^2 decomposition (Fig. 7) shows that beyond roughly five days, the perturbation method explains only a small fraction of variance in CRPS and ensemble spread relative to model choice. This is consistent with a growing role of model-dependent error at longer lead times (due to autoregressive error accumulation), plausibly arising from architectural bias, loss formulation, and underrepresented physical feedbacks. Combining input perturbations with model-checkpoint ensembles has been shown to yield higher probabilistic skill [Mahesh et al., 2025a, Baño-Medina et al., 2025b], but producing multiple checkpoints or very large ensembles is computationally expensive. Alternative approaches to introducing model stochasticity via latent-space, generative perturbations, or multi-model ensembles remain a promising research direction.

From a practical standpoint, the similarity in probabilistic skill between state-independent (Gaussian, Perlin) and flow-dependent (HCBV, HENS) perturbations is important considering the computational cost. Bred-vector methods require additional forward passes ($d = 3$ breeding cycles in our configuration), whereas Gaussian and Perlin perturbations are essentially free. Given that these simpler methods match or slightly exceed flow-dependent approaches on global metrics, they offer the most favorable cost trade-off for deterministic AIWP models, at least at 1° resolution and 10-day horizons. Ensemble size is a second practical lever: the decrease of importance of perturbation choice on spread and the dominance of model choice beyond day 5 suggest that smaller ensembles may suffice for short-range applications, with larger ensembles reserved for tail discrimination [Mahesh et al., 2025a, Baño-Medina et al., 2025b]. An ensemble size sensitivity analysis (Fig. 11) revealed that AIFS ENS dominates the skill–compute trade-off across all variables and ensemble sizes. However, AIFS ENS requires 38 GB of GPU memory by default [ECMWF, 2026]. On hardware with less memory the latency or run time will be higher. At short lead times, for 2m temperature, FuXi and GraphCast with Gaussian or Perlin perturbations approach (though do not reach) AIFS ENS skill with 30 ensemble members at a substantially lower memory footprint. A systematic study of the skill–cost frontier would be valuable future work. Input perturbations can be useful when native probabilistic models are not available, or when memory or latency is the binding constraint, especially at short lead times.

A broader pattern emerges from our results: spatially correlated noise, when appropriately scaled, appears sufficient to generate skillful AIWP ensembles regardless of its specific structure. Gaussian and Perlin perturbations differ substantially in their spectral properties (Appendix D) yet yield spread and skill comparable to flow-dependent bred vectors, suggesting that spatial coherence matters more than alignment with local flow. In Baño-Medina et al. [2025b], they describe how white noise (spatially uncorrelated) perturbations eventually map onto the growing modes of the forecast, after some autoregressive steps. Our results seem to suggest that providing spatially correlated noise is sufficient to allow the projection to happen in the first forecast step. This is however variable-dependent: FuXi and GraphCast ensembles match or exceed ENS on global temperature CRPS up to 3–5 days, but precipitation skill degrades more quickly. Predictive performance therefore depends not only on event rarity but also on the physical nature of the target [Zhao et al., 2025]. Temperature extremes are often spatially coherent and governed by persistent synoptic features [Yang et al., 2020], which AIWP models capture well; precipitation extremes, in contrast, depend on subgrid processes underrepresented in training data [Hess and Boers, 2022]. Improving AIWP generalization for rare hydro-meteorological extremes may therefore require targeted loss weighting, balanced sampling, or training with precipitation-specific datasets [Sun et al., 2025, Hess and Boers, 2022].

Model performance also depends on feature representation rather than the region alone: FuXi captures the China heatwave well, but underperforms on the monsoon-driven Pakistan precipitation. Evaluating AIWP ensembles on a feature basis linked to atmospheric regimes or teleconnection patterns could yield more diagnostic insight than regional averages. This study is further limited in time scale, geography, and resolution; extensions to seasonal variability, other regions, and higher resolutions (given that 1° may penalize precipitation performance) are natural next steps [Hess and Boers, 2022].

Overall, global metrics and case studies give complementary views: globally, perturbation-based AIWP ensembles are broadly competitive on average, while case studies expose where that skill translates into useful discrimination of high-impact events. Deterministic AI weather models can approximate aspects of probabilistic behavior through input perturbations such as simple spatially coherent noise and bred-vector approaches, but still fall short of fully capturing uncertainty and extremes. Gaussian and Perlin methods offer a promising bridge toward more realistic ensemble forecasts at low computational cost. Future work could integrate spatially coherent perturbations with generative or latent-space stochastic approaches, improve training data for extremes, or explore multi-model ensembles.

7 Conclusions

This study evaluated how deterministic AIWP models respond to input perturbations and how their resulting ensembles represent uncertainty during extreme events. Using FuXi, GraphCast, and SFNO, we generated 50-member ensembles with Gaussian, Perlin, HCBV, and HENS perturbations, evaluated globally over August 2022 and on the 2022 Pakistan floods and China heatwave case studies. To our knowledge, this is the first multi-model, multi-perturbation comparison of ensemble generation strategies for deterministic AIWP models that combines case-study and global threshold analyses.

Our central finding is that the dominant factor governing ensemble skill is the underlying model architecture, while the choice of perturbation method plays a secondary role: simpler methods produce similarly skillful ensembles to flow-dependent ones when appropriately scaled. Perturbation-based ensembles narrow but do not close the gap to NWP (ENS) or to native probabilistic AIWP models such as AIFS ENS, which remains the recommended choice where hardware permits. Deterministic AIWP models with simple perturbations are most attractive when memory or latency is a constraint. Across variables, all AIWP models captured temperature extremes more reliably than precipitation, and probabilistic skill declined with lead time.

Overall, perturbation-based ensembles extend deterministic AI models toward operational probabilistic forecasting. However, fully capturing the uncertainty and multiscale dynamics of extreme events could require hybrid strategies that integrate input perturbations with latent-space perturbations and generative approaches, improved training on rare extremes, or multi-model ensemble frameworks. Advancing along these directions will be key for building trustworthy, AI-driven early warning systems for high-impact weather events.

Acknowledgments

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Data availability statement

The ERA5 data in this study was obtained using the ERA5 ARCO dataset hosted on Google Cloud `https://github.com/google-research/arco-era5`. The IFS ENS data in this study was obtained from the WeatherBench 2 dataset `https://weatherbench2.readthedocs.io/en/latest/data-guide.html`. The code for processing the data, running the perturbations, and computing the metrics can be found in `https://gitlab.hhi.fraunhofer.de/ai-aml/aiwpuq`.

A Evaluation Metrics, Statistical Significance Testing and Variance Decomposition

For the following metric definitions, y_{nt} is the ERA5 value for each spatial location n (N spatial locations in a study region) at time $t \in \{1, \ldots, T\}$, with T as the number of forecast timesteps, and $\hat{y}_{nt}^{(m)}$ as the forecast provided by an ensemble member $m \in \{1, \ldots, M\}$, where $M = 50$.

A.1 Root Mean Squared Error (RMSE)

The Root Mean Squared Error (RMSE) is a metric that measures the average magnitude of errors between predicted and observed values [Rasp et al., 2024]. It penalizes larger errors more heavily, making it sensitive to outliers. Lower RMSE represents a better forecast. RMSE is defined as follows:

$$\mathrm{RMSE} = \sqrt{\frac{1}{NT} \sum_{n=1}^{N} \sum_{t=1}^{T} \left(y_{nt} - \frac{1}{M} \sum_{m=1}^{M} \hat{y}_{nt}^{(m)} \right)^2} \tag{5}$$

A.2 Continuous Ranked Probability Score (CRPS)

The Continuous Ranked Probability Score (CRPS) is a scoring rule that measures the accuracy of probabilistic forecasts by quantifying the difference between the predicted cumulative distribution function (CDF) and the observed outcome [Rasp et al., 2024]. It generalizes the mean absolute error to probabilistic forecasts. Lower CRPS indicates better forecast skill. CRPS is defined as follows:

$$\mathrm{CRPS} = \frac{1}{NT} \sum_{n=1}^{N} \sum_{t=1}^{T} \left[\frac{1}{M} \sum_{m=1}^{M} \left| \hat{y}_{nt}^{(m)} - y_{nt} \right| - \frac{1}{2M^2} \sum_{m=1}^{M} \sum_{k=1}^{M} \left| \hat{y}_{nt}^{(m)} - \hat{y}_{nt}^{(k)} \right| \right] \tag{6}$$

A.3 Receiver Operating Characteristic Skill Score (ROCSS)

The Receiver Operating Characteristic Skill Score (ROCSS) is a metric derived from the Area Under the ROC Curve (AUC) that quantifies the ability of a forecasting system to discriminate between the occurrence and non-occurrence of an event, relative to random chance [Zhao et al., 2025].

We set the event threshold τ to the 99th climatological percentile from ERA5 (1990–2020) to capture the extremes relevant to this study. The 2022 Pakistan floods were a record-breaking precipitation event [Singh et al., 2024], warranting a high percentile threshold to isolate tail behavior. For heat extremes in China, the 99th percentile aligns with established heatwave-impact studies such as Sun et al. [2021], where severe heatwave conditions are characterized using upper-percentile exceedance. Together, these justify the choice of $\tau = P_{99}$ as an appropriate discriminator for rare, high-impact extremes.

The computation of the ROCSS proceeds as follows:

1. Thresholding observations and ensemble forecasts: Given the threshold τ, observations and ensemble forecasts are converted into binary outcomes. Then, ensemble binary forecasts are converted into forecast probabilities:

$$o_{nt} = \mathbf{1}\{y_{nt} > \tau\}, \quad f_{nt}^{(m)} = \mathbf{1}\{\hat{y}_{nt}^{(m)} > \tau\}, \quad r_{nt} = \sum_{m=1}^{M} f_{nt}^{(m)}, \quad p_{nt} = \frac{r_{nt}}{M+1} \tag{7}$$

   Here, o_{nt} is the binary observation, $f_{nt}^{(m)}$ is the binary forecast of ensemble member m, r_{nt} is the ensemble "vote count" and p_{nt} is the resulting forecast probability using Weibull's plotting position [Weibull, 1939].

2. Applying probability thresholds: To construct the ROC curve, the forecast probability is converted back to binary outcomes for each probability threshold $\theta \in [0, 1]$:

$$\hat{o}_{nt}(\theta) = \mathbf{1}\{p_{nt} \geq \theta\} \tag{8}$$

3. Calculating Hit Rate (HR) and False Alarm Rate (FAR):
$$\mathrm{HR}(\theta) = \frac{\sum_{n,t} \mathbf{1}\{\hat{o}_{nt}(\theta) = 1 \wedge o_{nt} = 1\}}{\sum_{n,t} o_{nt}}, \qquad \mathrm{FAR}(\theta) = \frac{\sum_{n,t} \mathbf{1}\{\hat{o}_{nt}(\theta) = 1 \wedge o_{nt} = 0\}}{\sum_{n,t}(1 - o_{nt})} \tag{9}$$
4. Constructing the ROC curve: The ROC curve plots the trade-off between hits and false alarms across all probability thresholds:
$$\{(\mathrm{FAR}(\theta), \mathrm{HR}(\theta)) : \theta \in [0,1]\} \tag{10}$$
5. Computing the AUC:
$$AUC_{\text{forecast}} = \int_0^1 \mathrm{HR}(\mathrm{FAR}^{-1}(x))\, dx \tag{11}$$
where $\mathrm{FAR}^{-1}(x)$ denotes the inverse mapping of the false alarm rate along the ROC curve.
6. Computing the ROCSS: Finally, the ROCSS compares the forecast AUC to a reference (random) forecast, which has $AUC_{\text{reference}} = 0.5$:
$$\mathrm{ROCSS} = \frac{AUC_{\text{forecast}} - AUC_{\text{reference}}}{1 - AUC_{\text{reference}}} \tag{12}$$
A higher ROCSS indicates better discriminatory skill, with 1 representing perfect skill and 0 corresponding to no skill.

A.4 Statistical Significance Testing

To assess whether the differences in model performance between AIWP model ensembles and ENS, specifically in ROCSS, are statistically significant, we employ a clustered panel t-test based on a panel regression framework, following Olivetti and Messori [2024].

For each variable (daily accumulated precipitation, TP_{24h}, and maximum daily temperature, $T2M_{24h}$) and forecast lead time, we first compute the difference in ROCSS between the AIWP model ensemble and ENS at each grid cell and validation date. The resulting dataset (diff) is structured as a panel with spatial and temporal dimensions, indexed by latitude and longitude coordinates and valid time:

$$\mathrm{diff}_{nt} = y_{nt} \tag{13}$$

where n denotes the spatial location (grid point) and t the time index (validation date). We then estimate the following panel model:

$$y_{nt} = \alpha + \varepsilon_{nt} \tag{14}$$

where α represents the mean difference in ROCSS between the AIWP model ensemble and ENS, and ε_{nt} is the error term. The model is fit with two-way clustered standard errors, accounting for correlation within both spatial entities (grid cells) and across time. From the fitted model, we extract the t-statistic associated with the intercept (α), representing whether the mean difference significantly differs from zero.

Significance thresholds are determined from the Student's t distribution with degrees of freedom equal to the number of panel observations. Differences are classified as statistically significant at the 5% level ($p < 0.05$) when the t-statistic exceeds these bounds. The `PanelOLS` implementation from the `linearmodels` Python package is used to perform the computations.

A.5 Variance Decomposition

To quantify the relative importance of perturbation method and model architecture in determining forecast performance, we perform a two-way analysis of variance (ANOVA) at each grid point and lead time. The two factors are the perturbation method P (Gaussian, Perlin, HCBV, HENS; $p = 4$ levels) and the model architecture M (FuXi, GraphCast, SFNO; $m = 3$ levels). For each combination of factors, the response variable Y_{ij} represents the value of a given metric (CRPS, RMSE, or ensemble spread) for perturbation i and model j. The total variability, or total sum of squares (SS) is decomposed into contributions from each factor:
$$SS_{\text{total}} = SS_{\text{perturbation}} + SS_{\text{model}} + SS_{\text{residual}} \tag{15}$$

where:

$$SS_{\text{perturbation}} = m \sum_{i=1}^{p} (\bar{Y}_{i\cdot} - \bar{Y}_{\cdot\cdot})^2, \qquad SS_{\text{model}} = p \sum_{j=1}^{m} (\bar{Y}_{\cdot j} - \bar{Y}_{\cdot\cdot})^2 \tag{16}$$

$$SS_{\text{residual}} = SS_{\text{total}} - SS_{\text{perturbation}} - SS_{\text{model}} \tag{17}$$

with $\bar{Y}_{i\cdot}$ the mean over all models for perturbation i, $\bar{Y}_{\cdot j}$ the mean over all perturbations for model j, and $\bar{Y}_{\cdot\cdot}$ the grand mean. The residual term captures the interaction between the perturbation method and model architecture, reflecting non-additive effects where certain perturbation–model combinations perform differently than expected from their individual contributions. Because the design has no replicates, the interaction effect and the residual error are not separately identifiable, so any unmodeled noise is absorbed into the interaction term.

The effect size for each component is quantified using η^2, defined as the fraction of total variance explained:

$$\eta^2_{\text{perturbation}} = \frac{SS_{\text{perturbation}}}{SS_{\text{total}}}, \qquad \eta^2_{\text{model}} = \frac{SS_{\text{model}}}{SS_{\text{total}}}, \qquad \eta^2_{\text{interaction}} = \frac{SS_{\text{residual}}}{SS_{\text{total}}}. \tag{18}$$

By construction, $\eta^2_{\text{perturbation}} + \eta^2_{\text{model}} + \eta^2_{\text{interaction}} = 1$. A value of $\eta^2 \approx 1$ for a given component indicates that it explains nearly all of the variance in the metric, while $\eta^2 \approx 0$ indicates negligible influence. The analysis is performed independently at each grid point and lead time. The stacked area plots in Fig. 7 show the area-weighted global mean of $\eta^2_{\text{perturbation}}$, η^2_{model}, and $\eta^2_{\text{interaction}}$ as a function of lead time. The spatial maps display the dominant factor at each grid point, defined as the component with the largest η^2, averaged across all lead times.

B Additional plots

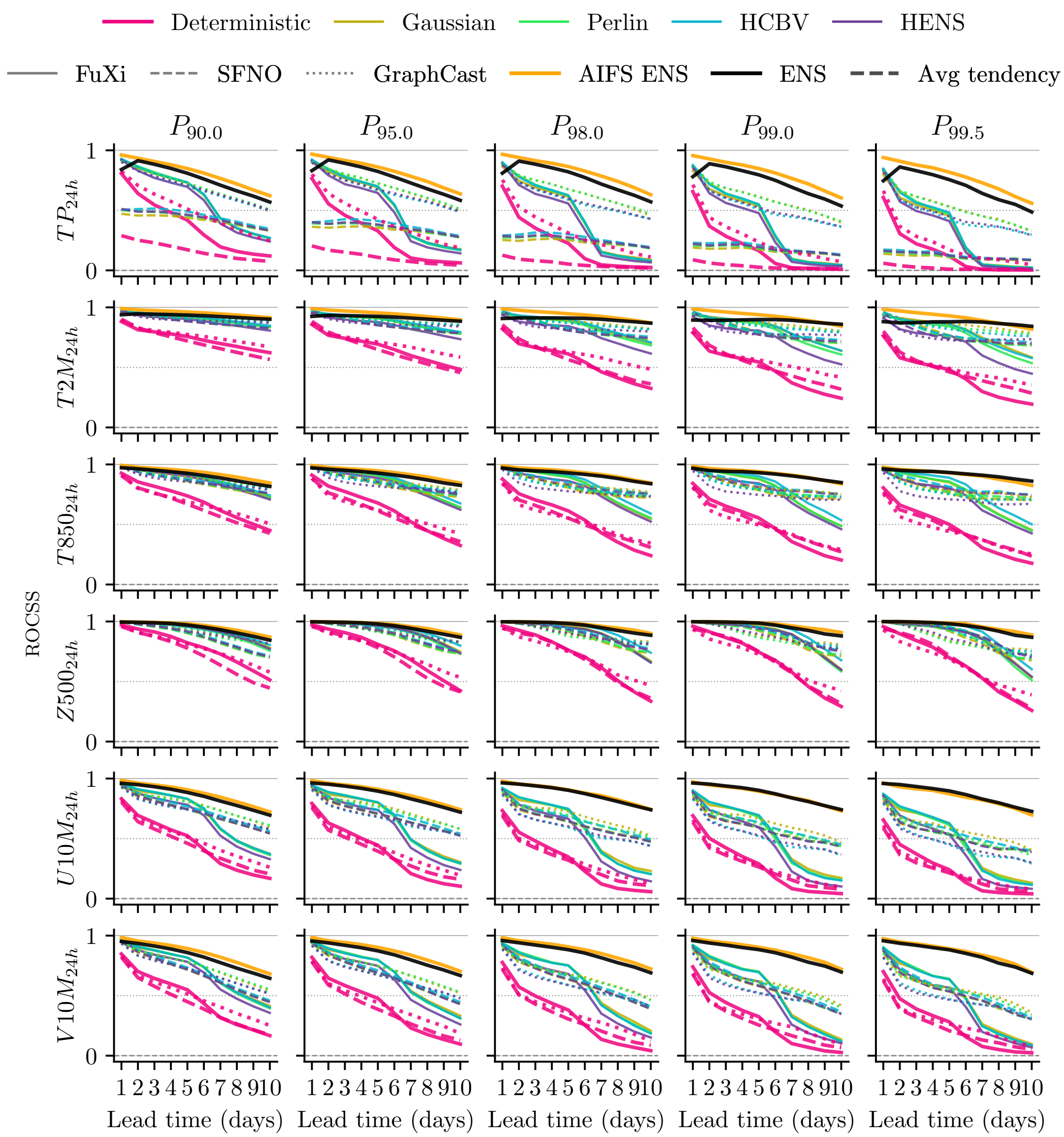

Figure 8: ROCSS values at the 90th till the 99.5th percentile of the ERA5 1990-2020 climatology for daily accumulated precipitation (TP_{24h}), maximum daily surface temperature ($T2M_{24h}$), temperature at 850hPa ($T850_{24h}$), geopotential at 500hPa ($Z500_{24h}$), and U and V 10 m height wind components ($U10M_{24h}$, $V10M_{24h}$), across 10 lead times, for the different AIWP models and their ensembles, globally. The higher the ROCSS values, the better the performance on the extremes.

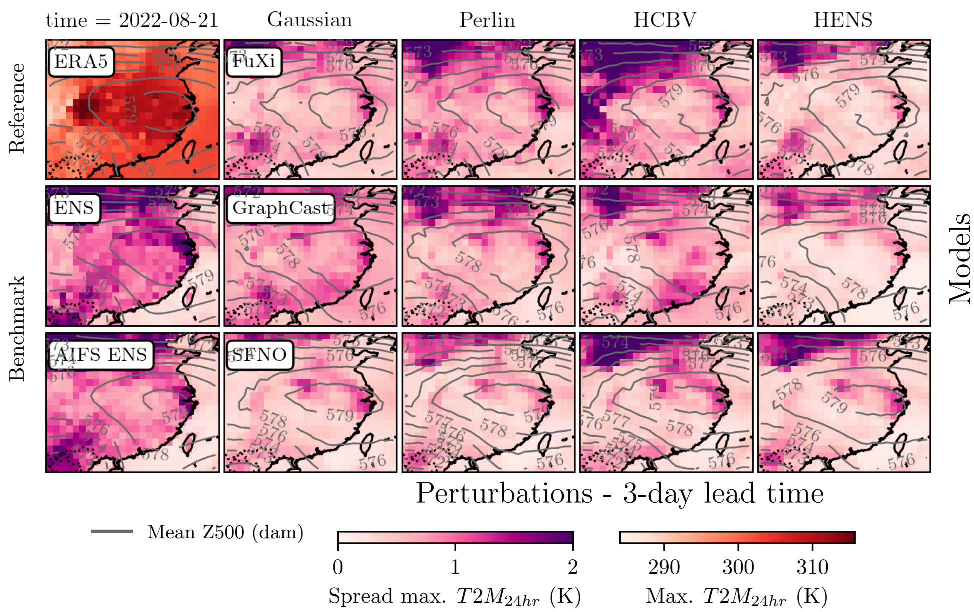

Figure 9: Daily maximum temperature spread for the different ensemble models (ENS, AIWPs) over China on 21st August 2022 for a 3-day lead time forecast. ERA5 spatial distribution for maximum temperature is shown in the top left corner. Daily average geopotential height at 500 hPa is also displayed as contour lines. AIFS ENS closely matches the ENS spread, while FuXi–Perlin comes closest among the perturbed ensembles.

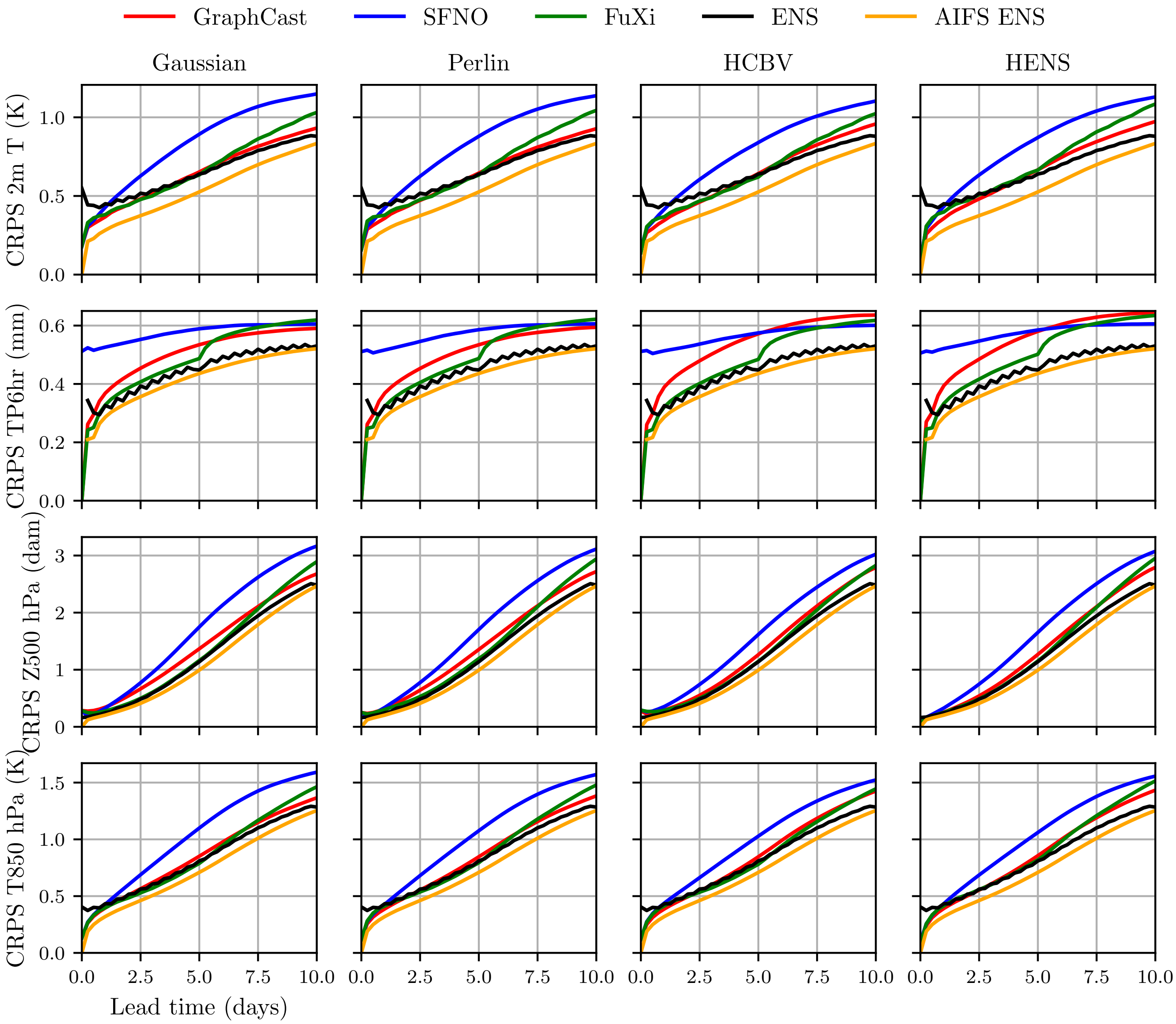

Figure 10: Global average CRPS over 10-day lead time for the different perturbation methods and AIWP models and ENS, in August 2022.

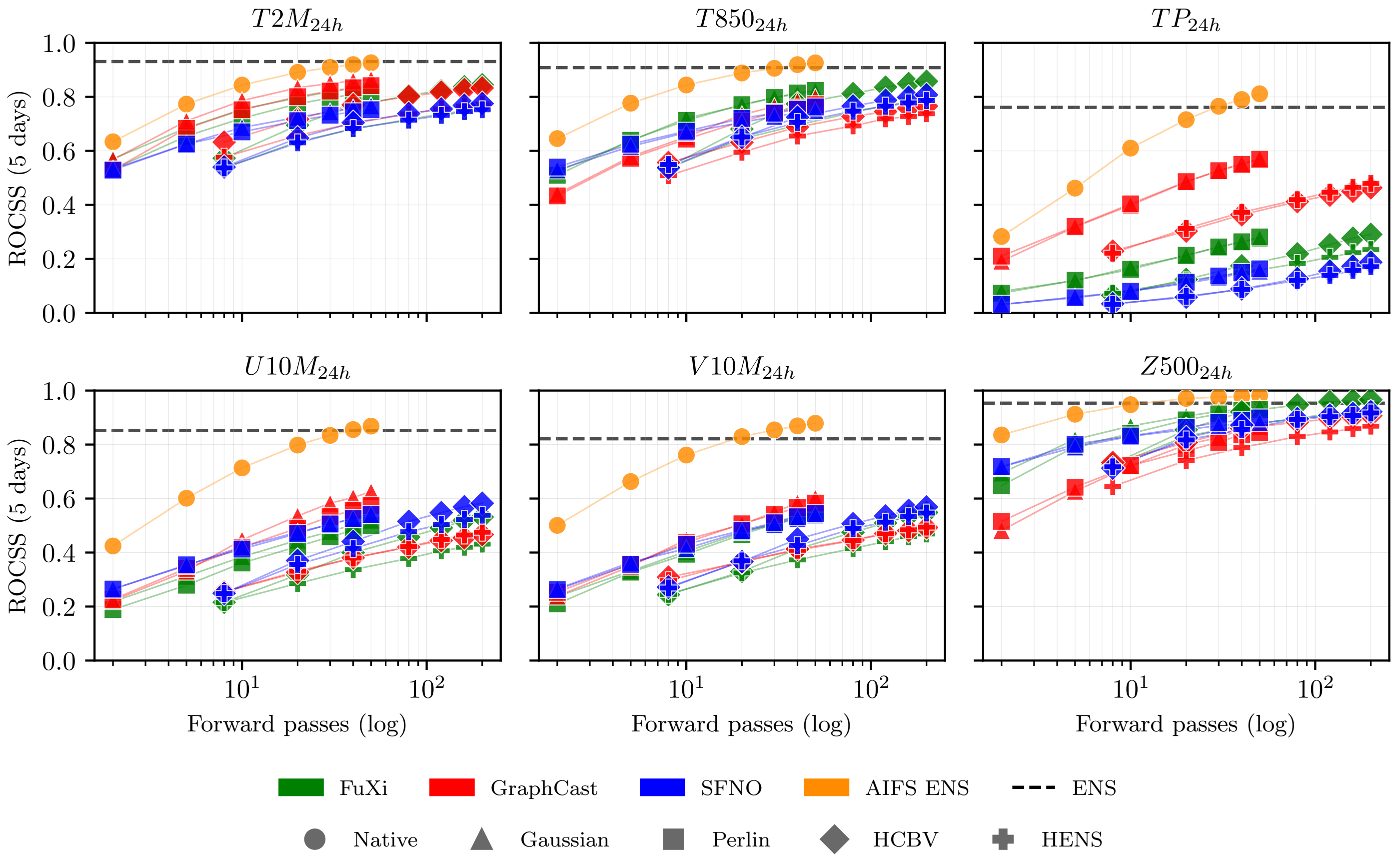

Figure 11: ROCSS for P_{99} and number of forward passes at 5 days lead time, for all six evaluation variables. Each marker corresponds to one (model, perturbation, ensemble size) configuration with color encoding the model and shape the perturbation method. Ensemble size is $N \in \{2, 5, 10, 20, 30, 40, 50\}$. For $N < 50$, ROCSS is computed on a single random sample of N members drawn from the full 50-member ensemble of each configuration. The x-axis reports the number of forward passes C required to produce the forecast, used as a model and hardware independent compute cost approximation: $C = N$ for Native (AIFS ENS), Gaussian, and Perlin perturbations, and $C = N\,(1 + d)$ with $d = 3$ breeding cycles per member for HCBV and HENS. The dashed line is the IFS ENS reference ROCSS. AIFS ENS dominates the skill–compute trade-off across all variables and ensemble sizes. For temperature at low budgets ($N = 30$), FuXi and GraphCast with Gaussian or Perlin are competitive with HCBV and HENS, and perform close to AIFS ENS, supporting the use of simple input perturbations when AIFS ENS is not available.

C Power Spectra of Forecasts

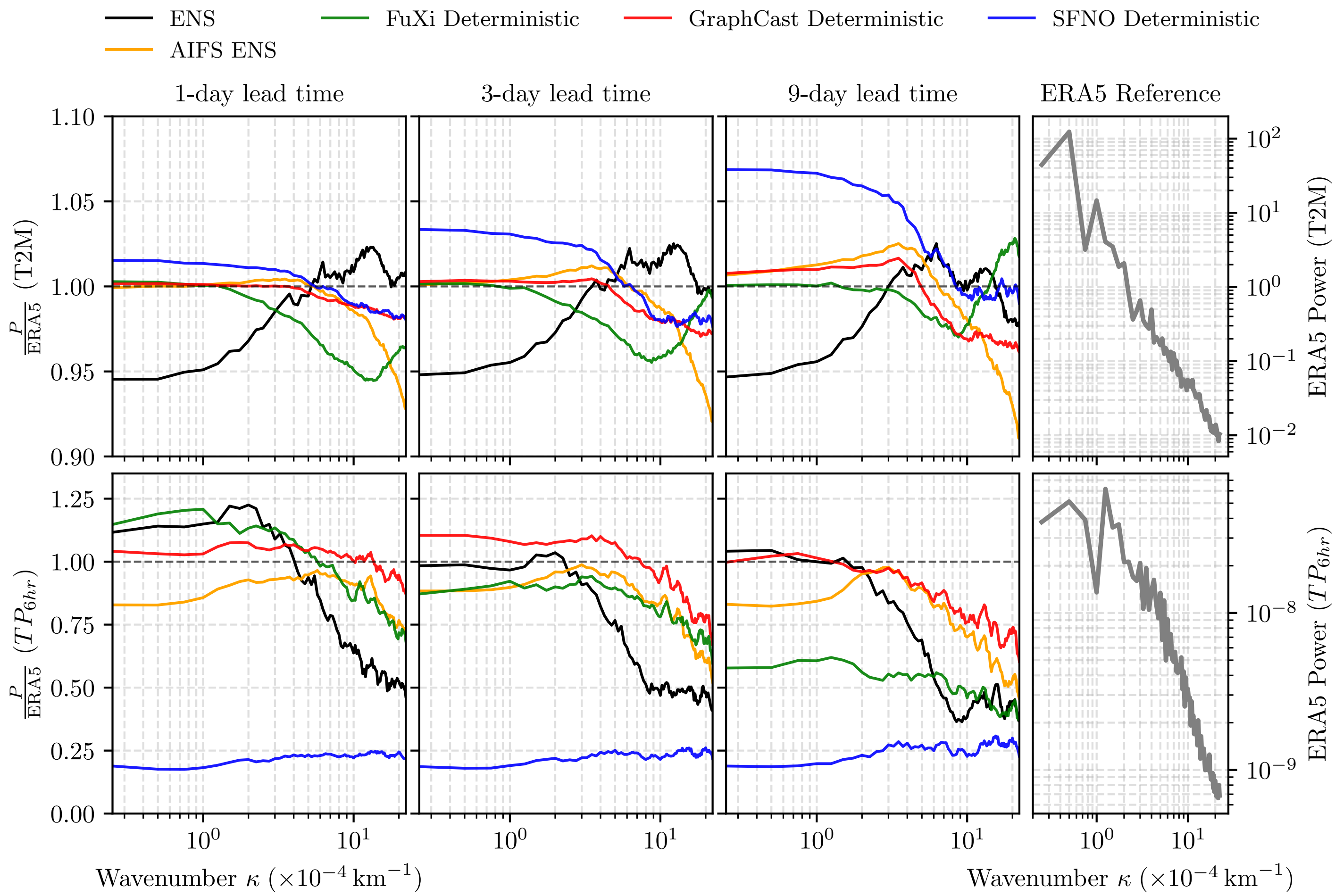

Figure 12: Spectral comparison of AIWP models against the ERA5 reanalysis for 2 m temperature (T2M, top row) and 6-hour accumulated precipitation (TP_{6h}, bottom row). Each panel shows the smoothed power ratio, $\frac{P}{\mathrm{ERA5}}$, as a function of wavenumber κ ($\times 10^{-4}\,\mathrm{km}^{-1}$), for the ensemble mean of ENS and AIFS ENS and the deterministic forecasts for FuXi, GraphCast, and SFNO, at forecast lead times of 1, 3, and 9 days. Values above 1 indicate excess variance relative to ERA5 (over-energetic spectra), while values under 1 indicate reduced small-scale variability (over-smoothing). The rightmost column displays the reference ERA5 power spectra for each variable.

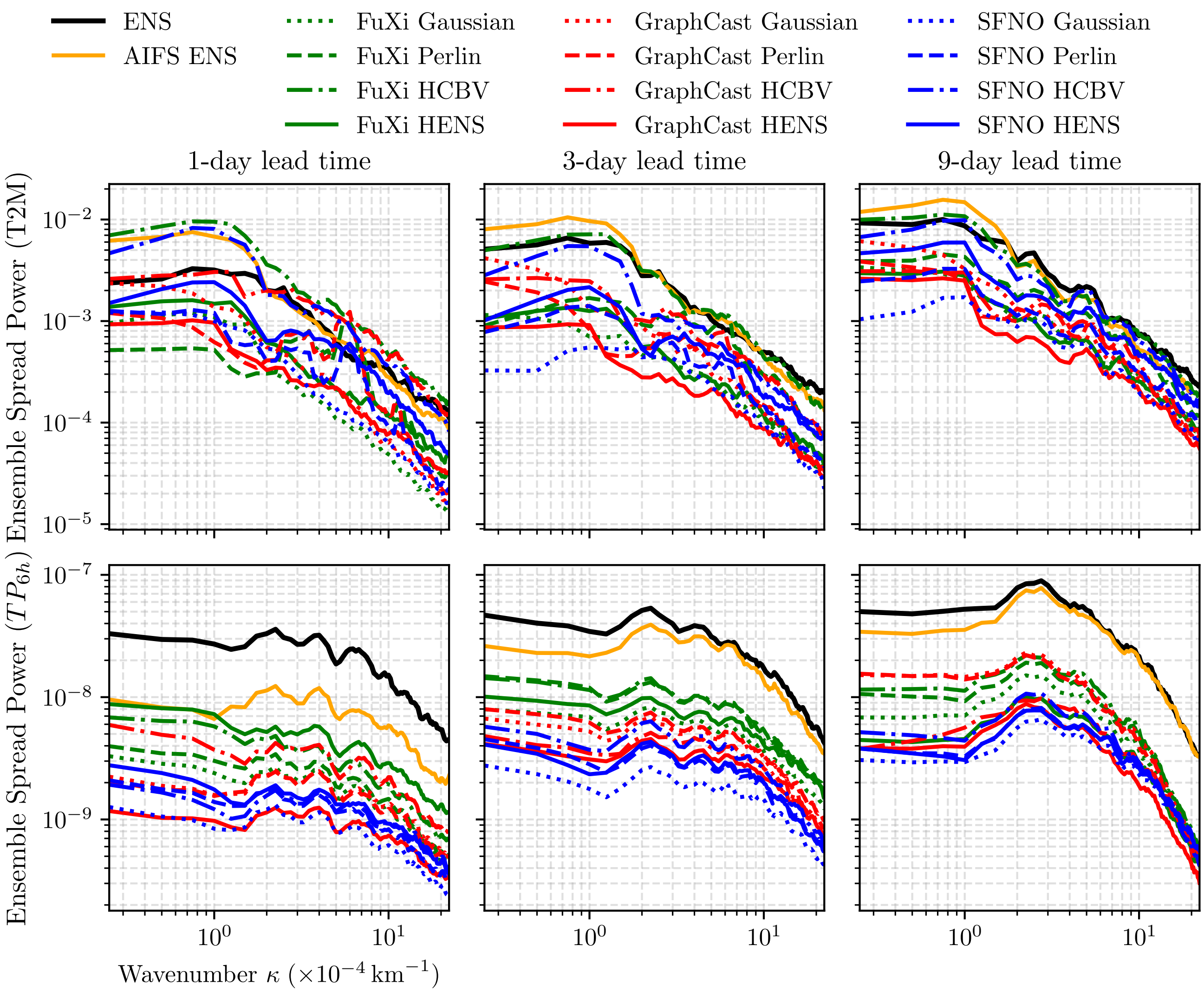

Figure 13: Ensemble spread spectral comparison for 2 m temperature (T2M, top row) and 6-hour accumulated precipitation (TP_{6h}, bottom row). Each panel shows the smoothed ensemble spread spectral power, as a function of wavenumber κ ($\times 10^{-4}\,\mathrm{km}^{-1}$), for ENS and AIFS ENS, and the perturbed forecasts for FuXi, GraphCast, and SFNO, at forecast lead times of 1, 3, and 9 days, for a single initialization time in August 15th, 2022.

D Visualization of perturbations

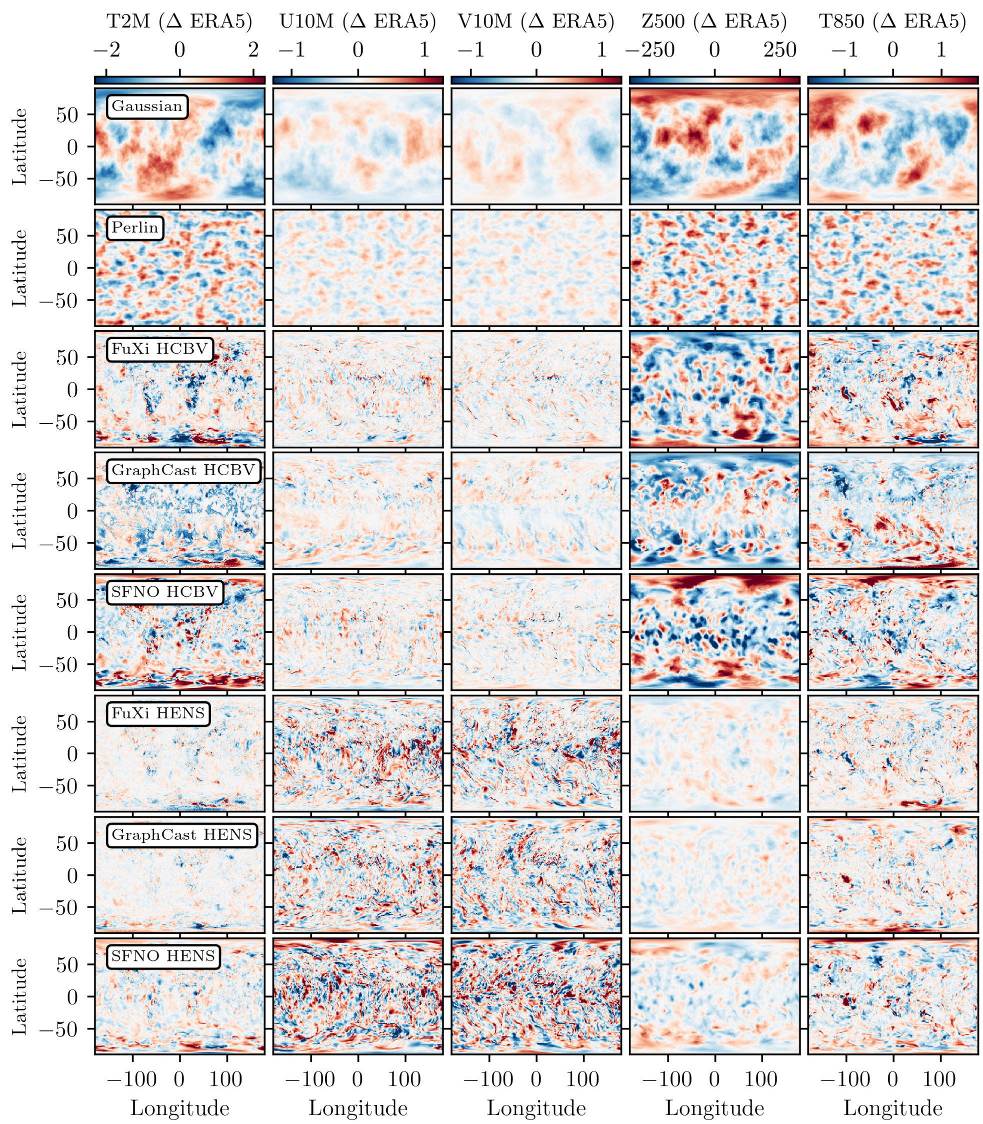

Figure 14: Visualization of the initial-condition perturbations applied to each AIWP model. Rows correspond to the different perturbation strategies: state-independent Gaussian and Perlin noise (shared across models), and the flow-dependent bred-vector methods HCBV and HENS, shown separately for FuXi, GraphCast, and SFNO. Columns correspond to the perturbed input variables, each displayed as a difference with respect to the ERA5 initial condition (ΔERA5), with red (blue) indicating positive (negative) values.

E Sensitivity analysis of the perturbation scaling factor

All four perturbation methods used in this study include a dimensionless scaling factor s that controls the overall magnitude of the perturbation applied to the initial state (Section 33.2). To select an appropriate value of s for each method, we performed a sensitivity analysis, described below.

We used a subset of our evaluation set for the scaling analysis: four initialization times on 15 August 2022 (00, 06, 12 and 18 UTC), 10 ensemble members per configuration, and a 5-day forecast rollout. The sweep was repeated for all three AIWP architectures (FuXi, GraphCast, SFNO) and all four perturbation methods (Gaussian, Perlin, HCBV, HENS). For Gaussian and HENS perturbations, s was varied directly over the values shown in Figs. 15 and 16. For HCBV, the effective scaling is one order of magnitude smaller than the value plotted on the axis (e.g., an axis value of 0.35 corresponds to an applied factor of 0.035); for Perlin, the effective scaling is one quarter of the axis value (e.g., 0.35 corresponds to 0.0875).

For each combination of perturbation method, model, and variable, we computed (i) the mean CRPS across lead times, (ii) the ensemble mean RMSE, and (iii) the ensemble standard deviation (spread). A good scaling factor should minimize CRPS and RMSE while producing a spread comparable to the ensemble error, in line with standard spread-skill consistency requirements. Across variables and models, we found that an axis value of $s = 0.35$ best satisfies these criteria for all four methods (corresponding to applied factors of 0.35 for Gaussian and HENS, 0.035 for HCBV, and 0.0875 for Perlin). This value is also broadly consistent with the scaling adopted by Mahesh et al. [2025a] for HENS. All results reported use these settings.

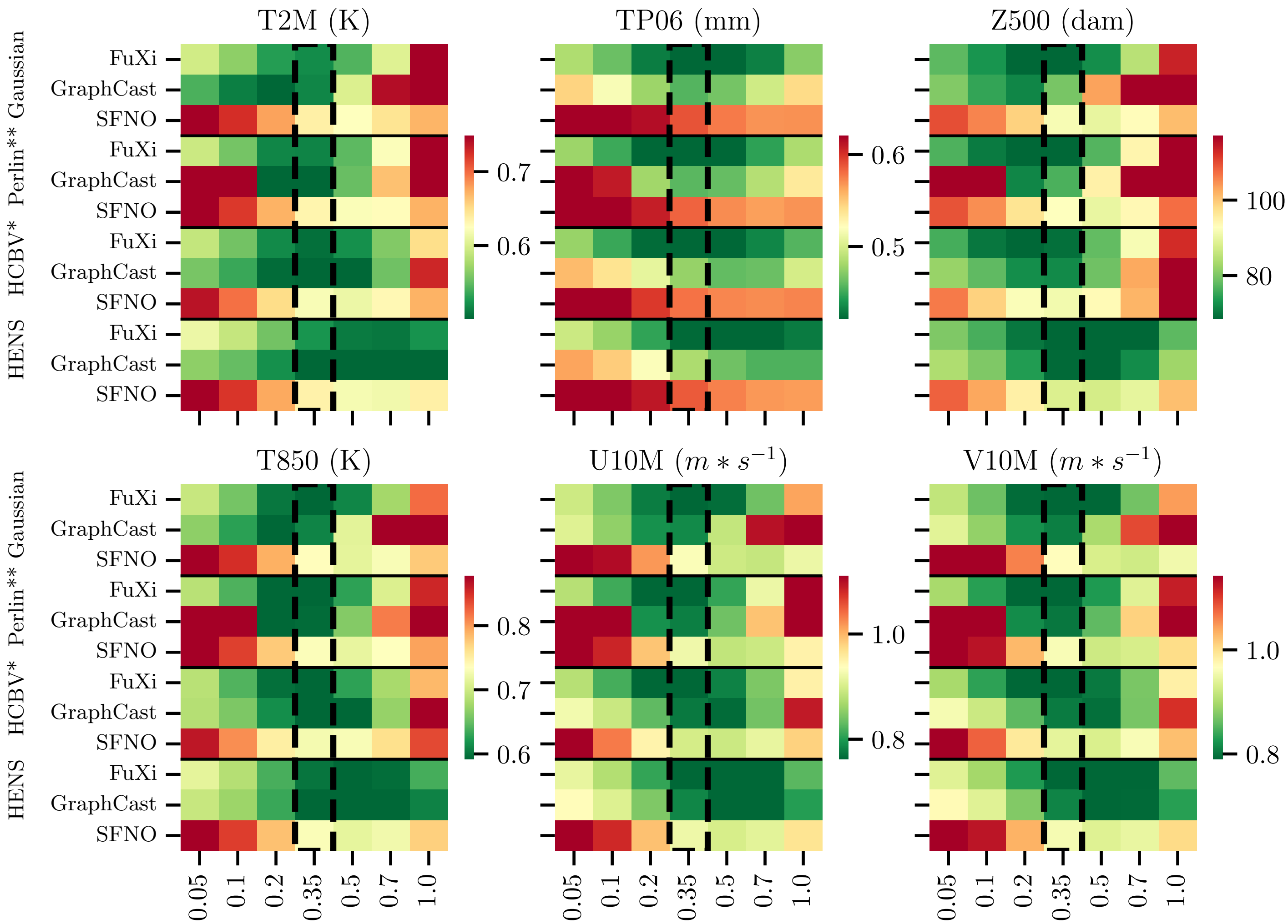

Figure 15: Mean CRPS across lead times as a function of the perturbation scaling factor s, for each combination of variable ($T2M$, TP_{6h}, $Z500$, $T850$, $U10M$, $V10M$), perturbation method (Gaussian, Perlin, HCBV, HENS), and the AIWP model (FuXi, GraphCast, SFNO). Lower CRPS indicates better probabilistic skill. For HCBV, the effective scaling factor is one order of magnitude smaller than the value shown on the axis (e.g., an axis value of 0.35 corresponds to an applied factor of 0.035); for Perlin, the effective scaling factor is one quarter of the axis value (e.g., 0.35 corresponds to 0.0875). The vertical dashed line marks the selected axis value of $s = 0.35$, which minimizes CRPS across most variable–model combinations and was used for all results reported in the main text.

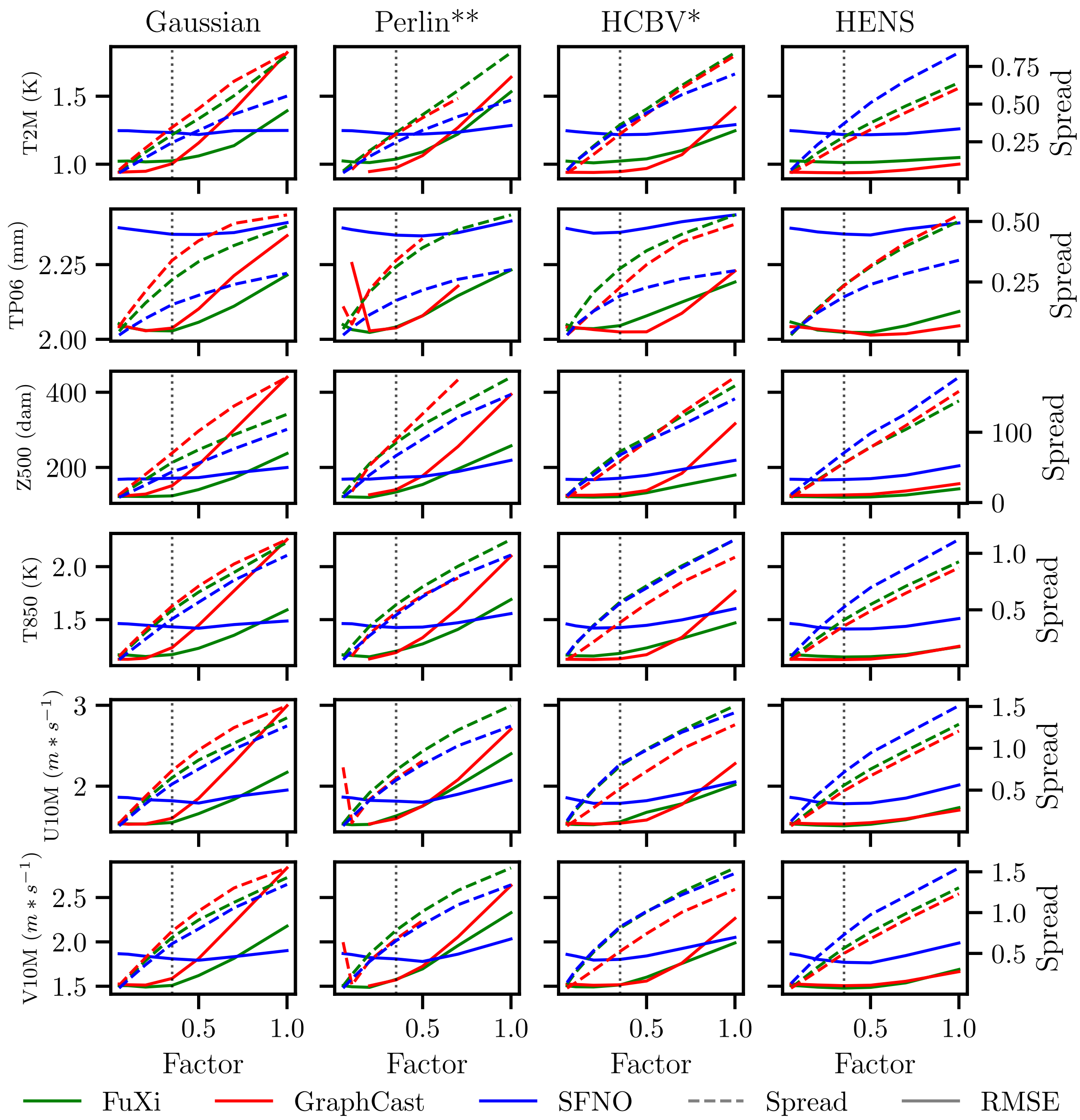

Figure 16: Ensemble standard deviation (spread, dashed) and ensemble-mean RMSE (solid) as a function of the perturbation scaling factor s, for each variable and perturbation method, with one curve per AIWP model (FuXi, GraphCast, SFNO). For HCBV, the effective scaling factor is one order of magnitude smaller than the value shown on the axis; for Perlin, it is one quarter of the axis value. The vertical dotted line marks the selected axis value of $s = 0.35$, which maximizes spread without increasing RMSE, across variable–model combinations.

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